



Flexi Mock CAT - 14 (2020)

Scorecard ([procreview.jsp?sid=aaaXgW7Hhf4V9fV8StDExSun Feb 14 15:57:52 IST 2021&qsetId=Rr8nceD7X0=&qsetName=Flexi Mock CAT - 14 \(2020\)\)](procreview.jsp?sid=aaaXgW7Hhf4V9fV8StDExSun Feb 14 15:57:52 IST 2021&qsetId=Rr8nceD7X0=&qsetName=Flexi Mock CAT - 14 (2020))))

Accuracy ([AccSelectGraph.jsp?sid=aaaXgW7Hhf4V9fV8StDExSun Feb 14 15:57:52 IST 2021&qsetId=Rr8nceD7X0=&qsetName=Flexi Mock CAT - 14 \(2020\)\)](AccSelectGraph.jsp?sid=aaaXgW7Hhf4V9fV8StDExSun Feb 14 15:57:52 IST 2021&qsetId=Rr8nceD7X0=&qsetName=Flexi Mock CAT - 14 (2020))))

Qs Analysis ([QsAnalysis.jsp?sid=aaaXgW7Hhf4V9fV8StDExSun Feb 14 15:57:52 IST 2021&qsetId=Rr8nceD7X0=&qsetName=Flexi Mock CAT - 14 \(2020\)\)](QsAnalysis.jsp?sid=aaaXgW7Hhf4V9fV8StDExSun Feb 14 15:57:52 IST 2021&qsetId=Rr8nceD7X0=&qsetName=Flexi Mock CAT - 14 (2020))))

Video Attempt ([VideoAnalysis.jsp?sid=aaaXgW7Hhf4V9fV8StDExSun Feb 14 15:57:52 IST 2021&qsetId=Rr8nceD7X0=&qsetName=Flexi Mock CAT - 14 \(2020\)\)](VideoAnalysis.jsp?sid=aaaXgW7Hhf4V9fV8StDExSun Feb 14 15:57:52 IST 2021&qsetId=Rr8nceD7X0=&qsetName=Flexi Mock CAT - 14 (2020))))

Solutions ([Solution.jsp?sid=aaaXgW7Hhf4V9fV8StDExSun Feb 14 15:57:52 IST 2021&qsetId=Rr8nceD7X0=&qsetName=Flexi Mock CAT - 14 \(2020\)\)](Solution.jsp?sid=aaaXgW7Hhf4V9fV8StDExSun Feb 14 15:57:52 IST 2021&qsetId=Rr8nceD7X0=&qsetName=Flexi Mock CAT - 14 (2020))))

Bookmarks ([Bookmarks.jsp?sid=aaaXgW7Hhf4V9fV8StDExSun Feb 14 15:57:52 IST 2021&qsetId=Rr8nceD7X0=&qsetName=Flexi Mock CAT - 14 \(2020\)\)](Bookmarks.jsp?sid=aaaXgW7Hhf4V9fV8StDExSun Feb 14 15:57:52 IST 2021&qsetId=Rr8nceD7X0=&qsetName=Flexi Mock CAT - 14 (2020))))

VARC

DILR

QA

Sec 1

Direction for questions (1-5): Read the given passage and answer the questions that follow

Let's face it: we've all had second thoughts about language. Hardly a day goes by when we don't stumble over words, stagger into misunderstandings, or struggle with a double negative. It's a frightfully cumbersome way to express ourselves. If language is such a slippery medium, perhaps it is time to replace it with something more dependable. Why not cut out the middleman and connect brains directly? The idea is not new. As the American physicist and Nobel laureate Murray Gell-Mann mused in *The Quark and the Jaguar* (1994): 'Thoughts and feelings would be completely shared, with none of the selectivity or deception that language permits.'

It is useful to examine this view of language carefully, for it is quite alluring. Computer scientist Rajesh Rao and his team complain about how hard it can be to verbalise feelings or forms of knowledge even if they are introspectively available. On Twitter, Musk has described words as 'a very lossy compression of thought'. How frustrating to have such a rich mental life and be stuck with such poor resources for expressing it! But no matter how much we can sympathise with this view, it misses a few crucial insights about language. First, words are tools. They can be misplaced or misused like any tool, but they are often useful for what they've been designed to do: help us say just what we want to say, and no more. When we choose our words carefully, it is because we know that there is a difference between private worlds and public words. There had better be, since social life depends on it.

Second, and more subtly, this view sees language as merely a channel for information: just as the speaking tube has made way for the telephone, so language can be done away with if we connect brains directly. This overlooks that language is also an infrastructure for social action. Think of everyday conversations, in which we riff off on a theme, recruit others to do stuff, relate to those around us. We don't just spout information indiscriminately; we apportion our words in conversational turns and build on each other's contributions. Language in everyday use is less like a channel and more like a tango: a fluid interplay of moves in which people can act as one, yet also retain their individuality. In social interaction there is room, by design, for consent and dissent.

The difference with current concepts of brain-to-brain interfaces couldn't be greater. A transcranial magnetic pulse leaves no room for doubt, but none for deliberation either. Its effect is as immediate as it is involuntary. We can admire the sheer efficiency of this form of interaction, but we also have to admit that something is lost. A sense of agency and autonomy; and along with that, perhaps even a sense of self. Nor does this problem go away merely by upgrading bandwidth, as is Musk's ambition for Neuralink, his implantable brain-computer interface. The very possibility of social (as opposed to merely symbiotic) life depends on there being some separation of private worlds, along with powers to interact on our own terms. In other words, we need something like language in order to be human.

Q.1 [11594329]

The author cites Nobel laureate Murray Gell-Mann in order to:

-
- 1 ☐ To indicate that language can be selective or deceptive at times.
-
- 2 ☐ To show the possibility of a mechanism through which thoughts and feelings can be shared more comprehensively.
-
- 3 ☐ To demonstrate how Gell-Mann found the idea of sharing thoughts and feelings completely without using language attractive.
-

4 ☐ To show that the idea of eliminating language for expression of thoughts and connecting brains directly is not new.



Solution:

Correct Answer : 4

Your Answer : 4

The author cites Gell-man as an example to support his assertion that the idea that brains be connected directly is not new. Hence, 4. 1 and 2 focus on the content of what Gell-man said, and not on why the author cites him. 3 incorrectly connects the example of Gell-man to the beginning of the second paragraph. The author has cited examples of Ramesh Rao and Musk to support the assertion about “that view of language” being alluring.

 Answer key/Solution

Bookmark

FeedBack

Direction for questions (1-5): Read the given passage and answer the questions that follow

Let's face it: we've all had second thoughts about language. Hardly a day goes by when we don't stumble over words, stagger into misunderstandings, or struggle with a double negative. It's a frightfully cumbersome way to express ourselves. If language is such a slippery medium, perhaps it is time to replace it with something more dependable. Why not cut out the middleman and connect brains directly? The idea is not new. As the American physicist and Nobel laureate Murray Gell-Mann mused in *The Quark and the Jaguar* (1994): 'Thoughts and feelings would be completely shared, with none of the selectivity or deception that language permits.'

It is useful to examine this view of language carefully, for it is quite alluring. Computer scientist Rajesh Rao and his team complain about how hard it can be to verbalise feelings or forms of knowledge even if they are introspectively available. On Twitter, Musk has described words as 'a very lossy compression of thought'. How frustrating to have such a rich mental life and be stuck with such poor resources for expressing it! But no matter how much we can sympathise with this view, it misses a few crucial insights about language. First, words are tools. They can be misplaced or misused like any tool, but they are often useful for what they've been designed to do: help us say just what we want to say, and no more. When we choose our words carefully, it is because we know that there is a difference between private worlds and public words. There had better be, since social life depends on it.

Second, and more subtly, this view sees language as merely a channel for information: just as the speaking tube has made way for the telephone, so language can be done away with if we connect brains directly. This overlooks that language is also an infrastructure for social action. Think of everyday conversations, in which we riff off on a theme, recruit others to do stuff, relate to those around us. We don't just spout information indiscriminately; we apportion our words in conversational turns and build on each other's contributions. Language in everyday use is less like a channel and more like a tango: a fluid interplay of moves in which people can act as one, yet also retain their individuality. In social interaction there is room, by design, for consent and dissent.

The difference with current concepts of brain-to-brain interfaces couldn't be greater. A transcranial magnetic pulse leaves no room for doubt, but none for deliberation either. Its effect is as immediate as it is involuntary. We can admire the sheer efficiency of this form of interaction, but we also have to admit that something is lost. A sense of agency and autonomy; and along with that, perhaps even a sense of self. Nor does this problem go away merely by upgrading bandwidth, as is Musk's ambition for Neuralink, his implantable brain-computer interface. The very possibility of social (as opposed to merely symbiotic) life depends on there being some separation of private worlds, along with powers to interact on our own terms. In other words, we need something like language in order to be human.

Q.2 [11594329]

The author opines all of the following about language EXCEPT:

- 1 ☐ It compresses thoughts so that they lose some of their essence.
 - 2 ☐ It can be misunderstood at times.
 - 3 ☐ Some of its grammatical elements can be challenging.
 - 4 ☐ It can result in people expressing themselves in a confused manner.
-



Solution:

Correct Answer : 1

Your Answer : 1

 Answer key/Solution

Refer the second sentence of Para 1. The author implies 2, 3, and 4 in it. To stumble over your words means to speak in a stuttering, confused manner. 1 is roughly the opinion of Elon Musk, not the author.

Bookmark

FeedBack

Direction for questions (1-5): Read the given passage and answer the questions that follow

Let's face it: we've all had second thoughts about language. Hardly a day goes by when we don't stumble over words, stagger into misunderstandings, or struggle with a double negative. It's a frightfully cumbersome way to express ourselves. If language is such a slippery medium, perhaps it is time to replace it with something more dependable. Why not cut out the middleman and connect brains directly? The idea is not new. As the American physicist and Nobel laureate Murray Gell-Mann mused in *The Quark and the Jaguar* (1994): 'Thoughts and feelings would be completely shared, with none of the selectivity or deception that language permits.'

It is useful to examine this view of language carefully, for it is quite alluring. Computer scientist Rajesh Rao and his team complain about how hard it can be to verbalise feelings or forms of knowledge even if they are introspectively available. On Twitter, Musk has described words as 'a very lossy compression of thought'. How frustrating to have such a rich mental life and be stuck with such poor resources for expressing it! But no matter how much we can sympathise with this view, it misses a few crucial insights about language. First, words are tools. They can be misplaced or misused like any tool, but they are often useful for what they've been designed to do: help us say just what we want to say, and no more. When we choose our words carefully, it is because we know that there is a difference between private worlds and public words. There had better be, since social life depends on it.

Second, and more subtly, this view sees language as merely a channel for information: just as the speaking tube has made way for the telephone, so language can be done away with if we connect brains directly. This overlooks that language is also an infrastructure for social action. Think of everyday conversations, in which we riff off on a theme, recruit others to do stuff, relate to those around us. We don't just spout information indiscriminately; we apportion our words in conversational turns and build on each other's contributions. Language in everyday use is less like a channel and more like a tango: a fluid interplay of moves in which people can act as one, yet also retain their individuality. In social interaction there is room, by design, for consent and dissent.

The difference with current concepts of brain-to-brain interfaces couldn't be greater. A transcranial magnetic pulse leaves no room for doubt, but none for deliberation either. Its effect is as immediate as it is involuntary. We can admire the sheer efficiency of this form of interaction, but we also have to admit that something is lost. A sense of agency and autonomy; and along with that, perhaps even a sense of self. Nor does this problem go away merely by upgrading bandwidth, as is Musk's ambition for Neuralink, his implantable brain-computer interface. The very possibility of social (as opposed to merely symbiotic) life depends on there being some separation of private worlds, along with powers to interact on our own terms. In other words, we need something like language in order to be human.

Q.3 [11594329]

What does the author allude to when they say, "When we choose our words carefully, it is because we know that there is a difference between private worlds and public words."

1 ☐ Choosing words carefully is not a drawback of language, it is rather an imperative imposed upon us by societal norms.

2 ☐ A careful use of words reflects the choice to not disclose all our inner thoughts to others, rather than the inadequacy of resources for expression.

3 ☐ Choosing our words carefully shows our desire to have an active social life, apart from a private life, rather than showing a drawback of language.

4 ☐ A person's careful use of words simply reflects their maturity as they become more conscious of the difference between their private world and their social life.

**Solution:****Correct Answer : 2****Your Answer : 2**[🔍 Answer key/Solution](#)

Refer Para 2. The author says that words help us say what we want to say, and no more. This coupled with the statement in the question, makes 2 the answer. The author makes these statements while defending language, implying that a careful choice of words is not a 'drawback'. 1 is eliminated because it is a 'choice', not an imperative, and societal 'norms' haven't been discussed. 3 misses the point of choosing our words carefully. 'Active' social life hasn't been discussed. 4 is incorrect because of 'more' and because it is incomplete as it does not talk about language not being a drawback.

[Bookmark](#)[FeedBack](#)

Direction for questions (1-5): Read the given passage and answer the questions that follow

Let's face it: we've all had second thoughts about language. Hardly a day goes by when we don't stumble over words, stagger into misunderstandings, or struggle with a double negative. It's a frightfully cumbersome way to express ourselves. If language is such a slippery medium, perhaps it is time to replace it with something more dependable. Why not cut out the middleman and connect brains directly? The idea is not new. As the American physicist and Nobel laureate Murray Gell-Mann mused in *The Quark and the Jaguar* (1994): 'Thoughts and feelings would be completely shared, with none of the selectivity or deception that language permits.'

It is useful to examine this view of language carefully, for it is quite alluring. Computer scientist Rajesh Rao and his team complain about how hard it can be to verbalise feelings or forms of knowledge even if they are introspectively available. On Twitter, Musk has described words as 'a very lossy compression of thought'. How frustrating to have such a rich mental life and be stuck with such poor resources for expressing it! But no matter how much we can sympathise with this view, it misses a few crucial insights about language. First, words are tools. They can be misplaced or misused like any tool, but they are often useful for what they've been designed to do: help us say just what we want to say, and no more. When we choose our words carefully, it is because we know that there is a difference between private worlds and public words. There had better be, since social life depends on it.

Second, and more subtly, this view sees language as merely a channel for information: just as the speaking tube has made way for the telephone, so language can be done away with if we connect brains directly. This overlooks that language is also an infrastructure for social action. Think of everyday conversations, in which we riff off on a theme, recruit others to do stuff, relate to those around us. We don't just spout information indiscriminately; we apportion our words in conversational turns and build on each other's contributions. Language in everyday use is less like a channel and more like a tango: a fluid interplay of moves in which people can act as one, yet also retain their individuality. In social interaction there is room, by design, for consent and dissent.

The difference with current concepts of brain-to-brain interfaces couldn't be greater. A transcranial magnetic pulse leaves no room for doubt, but none for deliberation either. Its effect is as immediate as it is involuntary. We can admire the sheer efficiency of this form of interaction, but we also have to admit that something is lost. A sense of agency and autonomy; and along with that, perhaps even a sense of self. Nor does this problem go away merely by upgrading bandwidth, as is Musk's ambition for Neuralink, his implantable brain-computer interface. The very possibility of social (as opposed to merely symbiotic) life depends on there being some separation of private worlds, along with powers to interact on our own terms. In other words, we need something like language in order to be human.

Q.4 [11594329]

Which of the following aspects of the current concepts of brain-to-brain interfaces makes them more like a "channel" and less like a "tango"?

1 ☐ Its efficiency

2 ☐ Its immediacy

3 ☐ Its involuntariness

4 ☐ Its fluidity



Solution:

Correct Answer : 3

Your Answer : 3

 Answer key/Solution

The author calls language more like a tango and less like a channel because it allows individuals to retain their individuality even after enabling them to act as one. The author then contrasts language with brain-to-brain interfaces. Here he/she points out how these can result in a loss of autonomy, agency and self. All this points to 3 as the answer. Fluidity is rejected as even language has been mentioned as a fluid interplay. Immediacy and efficiency, although features of these interfaces, are not the main points of contrast.

Bookmark

FeedBack

Direction for questions (1-5): Read the given passage and answer the questions that follow

Let's face it: we've all had second thoughts about language. Hardly a day goes by when we don't stumble over words, stagger into misunderstandings, or struggle with a double negative. It's a frightfully cumbersome way to express ourselves. If language is such a slippery medium, perhaps it is time to replace it with something more dependable. Why not cut out the middleman and connect brains directly? The idea is not new. As the American physicist and Nobel laureate Murray Gell-Mann mused in *The Quark and the Jaguar* (1994): 'Thoughts and feelings would be completely shared, with none of the selectivity or deception that language permits.'

It is useful to examine this view of language carefully, for it is quite alluring. Computer scientist Rajesh Rao and his team complain about how hard it can be to verbalise feelings or forms of knowledge even if they are introspectively available. On Twitter, Musk has described words as 'a very lossy compression of thought'. How frustrating to have such a rich mental life and be stuck with such poor resources for expressing it! But no matter how much we can sympathise with this view, it misses a few crucial insights about language. First, words are tools. They can be misplaced or misused like any tool, but they are often useful for what they've been designed to do: help us say just what we want to say, and no more. When we choose our words carefully, it is because we know that there is a difference between private worlds and public words. There had better be, since social life depends on it.

Second, and more subtly, this view sees language as merely a channel for information: just as the speaking tube has made way for the telephone, so language can be done away with if we connect brains directly. This overlooks that language is also an infrastructure for social action. Think of everyday conversations, in which we riff off on a theme, recruit others to do stuff, relate to those around us. We don't just spout information indiscriminately; we apportion our words in conversational turns and build on each other's contributions. Language in everyday use is less like a channel and more like a tango: a fluid interplay of moves in which people can act as one, yet also retain their individuality. In social interaction there is room, by design, for consent and dissent.

The difference with current concepts of brain-to-brain interfaces couldn't be greater. A transcranial magnetic pulse leaves no room for doubt, but none for deliberation either. Its effect is as immediate as it is involuntary. We can admire the sheer efficiency of this form of interaction, but we also have to admit that something is lost. A sense of agency and autonomy; and along with that, perhaps even a sense of self. Nor does this problem go away merely by upgrading bandwidth, as is Musk's ambition for Neuralink, his implantable brain-computer interface. The very possibility of social (as opposed to merely symbiotic) life depends on there being some separation of private worlds, along with powers to interact on our own terms. In other words, we need something like language in order to be human.

Q.5 [11594329]

Based on the features of the current brain-to-brain interfaces mentioned in the last paragraph, in which of the following scenarios would their use be the most suitable for communication?

-
- 1 ☐ A discussion between team members on how to proceed with a new project.
-
- 2 ☐ An exchange between a military commander and his junior in a war-like situation that demands clear, immediate, and carefully thought out commands.
-
- 3 ☐ An exchange between an art connoisseur and a painter where the connoisseur wants the painter to interpret and paint the connoisseur's unfiltered impulses.
-

4 ☐ An exchange between a couple that wants a quick and lasting resolution of their differences.



Solution:

Correct Answer : 3

Your Answer : 3

1 is rejected because a discussion about a new project would need deliberation and not hasty exchanges. 2 is rejected because it mentions that the commands have to be 'carefully thought'. 4 is rejected because the couple in order to arrive at a lasting resolution would want to deliberate, negotiate and both the participants would not want to lose their agency. 3 is the best choice as the connoisseur wants to convey his 'unfiltered impulses' to the painter, and words may distort them.

 Answer key/Solution

Bookmark

FeedBack

Direction for questions (6-9): Read the given passage and answer the questions that follow.

Coronavirus and political anxiety have accelerated this shift toward comfort TV. Who wants to be frustrated or made sadder by their entertainment when the outside world itself feels like a sufficiently brutal place?

On the other side stands Prestige TV -- shows watched for the "prestige" they carry -- that has always been difficult to categorize, in part because it is a mirage. Its gradual dissolution over the past several years is equally tricky to nail to any specific benchmark. But since the fever pitch of critical excitement over shows like *Mad Men*, *The Wire*, and the early days of *Game of Thrones*, prestige TV as an aesthetic has persisted, even while prestige as a reliable indicator of exciting TV you must talk about has undeniably waned.

In 2013, *Vulture* published "The 13 Rules for Creating a Prestige TV Drama," a full-feature list that began with installing a middle-aged anti-hero at the end of some major cultural era, making him great at his job and bad at his family, providing a healthy dose of both sex and violence, and sprinkling in a smattering of highfalutin cultural references to make sure the audience could recognize the show's pedigree and intelligence. I wrote an updated version of that list in 2017 reflecting the way prestige has become as much about how we talk about the genre as the genre itself. In 2015, as a bouncing-off point to define a genre he called "mid-reputable TV," the critic Noel Murray laid out some "hallmarks of a prestige drama -- heavy themes, high production values, accomplished actors," before conceding that "prestige is a state of mind." Much of its definition comes from a show's reception, Murray concluded. "Prestige television is often subject to intense scrutiny, with fans and critics evaluating every plot twist, stylistic choice, and coded method (in terms of both literary symbolism and the show's attitudes about gender, race, and politics)." In an essay titled "How American TV Became Respectable Without Getting Better," Matthew Christman offers a compelling definition: "Mostly crime. Mostly male. Mostly extravagantly unlikable anti-heroes whose sheer awfulness makes us feel better about our own, more mundane foibles."

I suspect some of the reason for prestige TV's simultaneous rise is that prestige, as a term, neatly dodges the best-versus-favorite (quality vs enjoyment) problem. Rather than define television by the pretense of objective greatness, the word prestige, with its snobbish implications, was indicative of a perhaps unintentional truth about these shows. It is a classist assessment, one that implies inaccessibility, scarcity, but also widespread influence and importance. It's not easy or fluffy or fun because, somewhere deep in the American psyche, there's still a puritanical belief that fun things cannot also be serious. Prestige TV is demanding, something that creates a further classist implication about its audience. It imagines itself being viewed by educated, wealthy people with enough leisure hours to devote to epic run times and slow-burning, byzantine plots (it imagines this regardless of who may actually be watching). It's not a coincidence that the outlets that defined prestige TV are also the ones, like HBO, that historically required an expensive monthly premium-cable package.

Q.6 [11594329]

All of the following are present in the 2013 full-feature list of "13 Rules for Creating a prestige TV Drama", published by *Vulture*, EXCEPT:

- 1 ☐ Grandiose cultural references
 - 2 ☐ A healthy dose of violence
 - 3 ☐ A white, middle-aged anti-hero
 - 4 ☐ Protagonist portrayed as good at his job but bad at his family
-

**Solution:****Correct Answer : 3****Your Answer : 3****Refer the first few sentences of Para 3. It is not mentioned that the anti-hero is white. Hence 3. The other features are mentioned.**[🔍 Answer key/Solution](#)[Bookmark](#)[FeedBack](#)**Direction for questions (6-9):** Read the given passage and answer the questions that follow.

Coronavirus and political anxiety have accelerated this shift toward comfort TV. Who wants to be frustrated or made sadder by their entertainment when the outside world itself feels like a sufficiently brutal place?

On the other side stands Prestige TV – shows watched for the “prestige” they carry – that has always been difficult to categorize, in part because it is a mirage. Its gradual dissolution over the past several years is equally tricky to nail to any specific benchmark. But since the fever pitch of critical excitement over shows like *Mad Men*, *The Wire*, and the early days of *Game of Thrones*, prestige TV as an aesthetic has persisted, even while prestige as a reliable indicator of exciting TV you must talk about has undeniably waned.

In 2013, *Vulture* published “The 13 Rules for Creating a Prestige TV Drama,” a full-feature list that began with installing a middle-aged anti-hero at the end of some major cultural era, making him great at his job and bad at his family, providing a healthy dose of both sex and violence, and sprinkling in a smattering of highfalutin cultural references to make sure the audience could recognize the show’s pedigree and intelligence. I wrote an updated version of that list in 2017 reflecting the way prestige has become as much about how we talk about the genre as the genre itself. In 2015, as a bouncing-off point to define a genre he called “mid-reputable TV,” the critic Noel Murray laid out some “hallmarks of a prestige drama – heavy themes, high production values, accomplished actors,” before conceding that “prestige is a state of mind.” Much of its definition comes from a show’s reception, Murray concluded. “Prestige television is often subject to intense scrutiny, with fans and critics evaluating every plot twist, stylistic choice, and coded method (in terms of both literary symbolism and the show’s attitudes about gender, race, and politics).” In an essay titled “How American TV Became Respectable Without Getting Better,” Matthew Christman offers a compelling definition: “Mostly crime. Mostly male. Mostly extravagantly unlikable anti-heroes whose sheer awfulness makes us feel better about our own, more mundane foibles.”

I suspect some of the reason for prestige TV’s simultaneous rise is that prestige, as a term, neatly dodges the best-versus-favorite (quality vs enjoyment) problem. Rather than define television by the pretense of objective greatness, the word prestige, with its snobbish implications, was indicative of a perhaps unintentional truth about these shows. It is a classist assessment, one that implies inaccessibility, scarcity, but also widespread influence and importance. It’s not easy or fluffy or fun because, somewhere deep in the American psyche, there’s still a puritanical belief that fun things cannot also be serious. Prestige TV is demanding, something that creates a further classist implication about its audience. It imagines itself being viewed by educated, wealthy people with enough leisure hours to devote to epic run times and slow-burning, byzantine plots (it imagines this regardless of who may actually be watching). It’s not a coincidence that the outlets that defined prestige TV are also the ones, like HBO, that historically required an expensive monthly premium-cable package.

Q.7 [11594329]

In Para 3, what does the author imply by the statement, "...prestige has become as much about how we talk about a genre as the genre itself."?

- 1 ☐ The audience considers prestige TV to be the TV that deals with heavy themes.
- 2 ☐ The categorisation of a show as prestige TV depends on the perception that a genre carries among the viewers.
- 3 ☐ Prestige TV is difficult to categorise because it does not just depend on the genre to which it belongs, but also on the viewers' state of mind.
- 4 ☐ The categorisation of a show as prestige TV is determined by the discussion that it generates on the genre to which it belongs.

**Solution:****Correct Answer : 2****Your Answer : 2**

'How we talk about a genre' here means how we perceive a genre. Hence 2. 1 is a fact stated by Murray, not implied by the author. 3 is incomplete, distorted and does not give the meaning of the phrase asked in the question. The last phrase should have been 'the viewers' perception of a genre'. 4 is eliminated because the sentence does not talk about the discussion "generated by the show" about the genre.

[🔍 Answer key/Solution](#)[Bookmark](#)[FeedBack](#)

Direction for questions (6-9): Read the given passage and answer the questions that follow.

Coronavirus and political anxiety have accelerated this shift toward comfort TV. Who wants to be frustrated or made sadder by their entertainment when the outside world itself feels like a sufficiently brutal place?

On the other side stands Prestige TV -- shows watched for the "prestige" they carry -- that has always been difficult to categorize, in part because it is a mirage. Its gradual dissolution over the past several years is equally tricky to nail to any specific benchmark. But since the fever pitch of critical excitement over shows like *Mad Men*, *The Wire*, and the early days of *Game of Thrones*, prestige TV as an aesthetic has persisted, even while prestige as a reliable indicator of exciting TV you must talk about has undeniably waned.

In 2013, *Vulture* published "The 13 Rules for Creating a Prestige TV Drama," a full-feature list that began with installing a middle-aged anti-hero at the end of some major cultural era, making him great at his job and bad at his family, providing a healthy dose of both sex and violence, and sprinkling in a smattering of highfalutin cultural references to make sure the audience could recognize the show's pedigree and intelligence. I wrote an updated version of that list in 2017 reflecting the way prestige has become as much about how we talk about the genre as the genre itself. In 2015, as a bouncing-off point to define a genre he called "mid-reputable TV," the critic Noel Murray laid out some "hallmarks of a prestige drama -- heavy themes, high production values, accomplished actors," before conceding that "prestige is a state of mind." Much of its definition comes from a show's reception, Murray concluded. "Prestige television is often subject to intense scrutiny, with fans and critics evaluating every plot twist, stylistic choice, and coded method (in terms of both literary symbolism and the show's attitudes about gender, race, and politics)." In an essay titled "How American TV Became Respectable Without Getting Better," Matthew Christman offers a compelling definition: "Mostly crime. Mostly male. Mostly extravagantly unlikable anti-heroes whose sheer awfulness makes us feel better about our own, more mundane foibles."

I suspect some of the reason for prestige TV's simultaneous rise is that prestige, as a term, neatly dodges the best-versus-favorite (quality vs enjoyment) problem. Rather than define television by the pretense of objective greatness, the word prestige, with its snobbish implications, was indicative of a perhaps unintentional truth about these shows. It is a classist assessment, one that implies inaccessibility, scarcity, but also widespread influence and importance. It's not easy or fluffy or fun because, somewhere deep in the American psyche, there's still a puritanical belief that fun things cannot also be serious. Prestige TV is demanding, something that creates a further classist implication about its audience. It imagines itself being viewed by educated, wealthy people with enough leisure hours to devote to epic run times and slow-burning, byzantine plots (it imagines this regardless of who may actually be watching). It's not a coincidence that the outlets that defined prestige TV are also the ones, like HBO, that historically required an expensive monthly premium-cable package.

Q.8 [11594329]

Which of the following, if true, would contribute the most to weakening Matthew Christman's views as mentioned in the passage?

- 1 ☐ American TV has given us some highly admired, principled protagonists or heroes over the years.
- 2 ☐ American TV has been very adaptable to responding to the needs and tastes of its viewers since its inception.
- 3 ☐ American TV, when analysed, revealed a cast that could be complimented for its equal sex ratio, with women playing roles that echoed the times.

4 ○ American TV is popular for the racial diversity that its shows have portrayed since its early days, with its eclectic mix going beyond African Americans.



Solution:

Correct Answer : 3

Your Answer : 3

Mathew Christman's essay calls American TV mostly crime, mostly male, mostly anti-hero. 1 is eliminated as Christman does not call the shows 'entirely' anti-hero, but 'mostly' anti-hero. 2 does not weaken the argument because Matthew indirectly acknowledges that American TV was 'respectable' and 'made people feel better about their own foibles'. It does not attack his main argument. 4 is eliminated because race has not been discussed by Matthew. 3 attacks his "mostly male" argument.

 Answer key/Solution

Bookmark

FeedBack

Direction for questions (6-9): Read the given passage and answer the questions that follow.

Coronavirus and political anxiety have accelerated this shift toward comfort TV. Who wants to be frustrated or made sadder by their entertainment when the outside world itself feels like a sufficiently brutal place?

On the other side stands Prestige TV -- shows watched for the "prestige" they carry -- that has always been difficult to categorize, in part because it is a mirage. Its gradual dissolution over the past several years is equally tricky to nail to any specific benchmark. But since the fever pitch of critical excitement over shows like *Mad Men*, *The Wire*, and the early days of *Game of Thrones*, prestige TV as an aesthetic has persisted, even while prestige as a reliable indicator of exciting TV you must talk about has undeniably waned.

In 2013, *Vulture* published "The 13 Rules for Creating a Prestige TV Drama," a full-feature list that began with installing a middle-aged anti-hero at the end of some major cultural era, making him great at his job and bad at his family, providing a healthy dose of both sex and violence, and sprinkling in a smattering of highfalutin cultural references to make sure the audience could recognize the show's pedigree and intelligence. I wrote an updated version of that list in 2017 reflecting the way prestige has become as much about how we talk about the genre as the genre itself. In 2015, as a bouncing-off point to define a genre he called "mid-reputable TV," the critic Noel Murray laid out some "hallmarks of a prestige drama -- heavy themes, high production values, accomplished actors," before conceding that "prestige is a state of mind." Much of its definition comes from a show's reception, Murray concluded. "Prestige television is often subject to intense scrutiny, with fans and critics evaluating every plot twist, stylistic choice, and coded method (in terms of both literary symbolism and the show's attitudes about gender, race, and politics)." In an essay titled "How American TV Became Respectable Without Getting Better," Matthew Christman offers a compelling definition: "Mostly crime. Mostly male. Mostly extravagantly unlikable anti-heroes whose sheer awfulness makes us feel better about our own, more mundane foibles."

I suspect some of the reason for prestige TV's simultaneous rise is that prestige, as a term, neatly dodges the best-versus-favorite (quality vs enjoyment) problem. Rather than define television by the pretense of objective greatness, the word prestige, with its snobbish implications, was indicative of a perhaps unintentional truth about these shows. It is a classist assessment, one that implies inaccessibility, scarcity, but also widespread influence and importance. It's not easy or fluffy or fun because, somewhere deep in the American psyche, there's still a puritanical belief that fun things cannot also be serious. Prestige TV is demanding, something that creates a further classist implication about its audience. It imagines itself being viewed by educated, wealthy people with enough leisure hours to devote to epic run times and slow-burning, byzantine plots (it imagines this regardless of who may actually be watching). It's not a coincidence that the outlets that defined prestige TV are also the ones, like HBO, that historically required an expensive monthly premium-cable package.

Q.9 [11594329]

What purpose does the last sentence of the last paragraph serve?

- 1 ☐ It presents a point of criticism of prestige TV.
 - 2 ☐ It sums up the author's criticism of prestige TV as classist and snobbish.
 - 3 ☐ It supports the assertion that outlets are making a lot of money by presenting prestige TV.
 - 4 ☐ It is an assertion that supports another assertion made earlier in the paragraph.
-



Solution:

Correct Answer : 4

Your Answer : 4

 Answer key/Solution

The last sentence supports the assertion that prestige TV is classist and imagines its audience to be wealthy, stated earlier in the paragraph. Hence, 4. 1 is eliminated as this assertion is a support of the criticism done earlier. 2 can be easily eliminated as the last sentence is not in the nature of a summary or conclusion. 3 is distorted because the sentence's focus is not on the outlets making a lot of money, but on their excluding people by requiring expensive packages, thus contributing to prestige TV's classist element.

Bookmark

FeedBack

Direction for questions (10-14): Read the given passage and answer the questions that follow.

I didn't know it at the time, but I was suffering from severe anxiety, a medical condition recognized by doctors going as far back as the ancient Greeks. Today, 11 different anxiety disorders are listed in the "Diagnostic and Statistical Manual of Mental Disorders," the bible for mental-health professionals, and they are being diagnosed in rising numbers.

Unfortunately, even as we have learned more about the varieties of anxiety in recent decades, the underlying causes of the disorder have remained poorly understood. But that is changing. Scientists have started to unravel some of the mysteries of the anxious brain, and their discoveries may soon lead to new and better treatments.

Still, rates of anxiety disorders, as well as depression, seem to be increasing among young people. The number of Americans expected to have at least one anxiety disorder in the course of their lives is staggering: one in three, ages 13 or older, according to researchers at Harvard Medical School and Technische Universität Dresden. The number is even higher for women—about 40%. Each year, about 40 million American adults have an anxiety disorder, and that doesn't include the millions of garden-variety worriers and insomniacs whose anxiety, though not debilitating, leaches away joy and steals their peace of mind.

Rates of anxiety disorders, as well as depression, seem to be increasing among young people, particularly college students. According to a 2016 survey by the American College Health Association, 17% of students were diagnosed with or treated for anxiety problems during the previous year, and nearly 14% were diagnosed with or treated for depression. That is up from about 10% each for anxiety and depression in 2008.

A certain amount of anxiety is a good thing, motivating us to study for tests, prepare for presentations, save for retirement. Too much anxiety, however, can be incapacitating and expensive. Anxiety disorders cost the U.S. about \$63 billion a year, according to a 1999 study published in the Journal of Clinical Psychiatry, the most recent estimate available. The tally includes doctor and hospital visits, psychiatric treatment, prescription drugs and the value of lost productivity at work. Anxiety can also lead to depression, substance abuse and even suicide. Anxious people who work have lower incomes. They are less likely to marry, and, if they do, more likely to divorce.

Scientists have found that many childhood events and experiences—from illness to trauma to overprotective and controlling styles of parenting—can contribute to the development of anxiety. It is also partly genetic. Having a first-degree relative—a parent, sibling or child—with an anxiety disorder bumps a person's risk of developing one by up to five times that of the general population.

I certainly don't see my anxiety as a gift, but it has some upsides. When I'm avoiding some necessary confrontation or saying yes to too many superfluous obligations, I feel it, and it kicks me into action. Weirdly, anxiety makes me live a more authentic life. And a more empathic one. It has made me feel vulnerable and more open to asking for help, thereby deepening my friendships.

People who have a brush with death often talk of how it has given them a sense of what really matters. An omnipresent fear of disaster and a constant bracing for catastrophe can do that, too. Time takes on more urgency. The background hum of uneasiness in my mind has motivated me to work harder, to speak more honestly and, curiously, to take more risks than I might otherwise have. Anxiety means that I'm simply not mellow enough to take things for granted. And that has made my life all the richer.

Q.10 [11594329]

The author has talked about people who have had a near death experience in order to:

- 1 ☐ argue that a certain amount of depression is a good thing.
- 2 ☐ illustrate that anxiety disorder has led him/her to acquire some positive traits.
- 3 ☐ prove that anxiety has made him/her richer.
- 4 ☐ illustrate that anxiety disorder is a blessing in disguise.

**Solution:**

Correct Answer : 2

Your Answer : 2

The author has spoken about this example in the last paragraph. This paragraph points out how anxiety disorder has made the author more purposeful and focused. 1 is wrong because the author is neither arguing nor talking about depression in the last paragraph. 3 is wrong because the word “richer” has been used figuratively in the passage. 4 is too vague to answer the question. Hence, 2 is the answer.

[🔍 Answer key/Solution](#)[Bookmark](#)[FeedBack](#)

Direction for questions (10-14): Read the given passage and answer the questions that follow.

I didn't know it at the time, but I was suffering from severe anxiety, a medical condition recognized by doctors going as far back as the ancient Greeks. Today, 11 different anxiety disorders are listed in the "Diagnostic and Statistical Manual of Mental Disorders," the bible for mental-health professionals, and they are being diagnosed in rising numbers.

Unfortunately, even as we have learned more about the varieties of anxiety in recent decades, the underlying causes of the disorder have remained poorly understood. But that is changing. Scientists have started to unravel some of the mysteries of the anxious brain, and their discoveries may soon lead to new and better treatments.

Still, rates of anxiety disorders, as well as depression, seem to be increasing among young people. The number of Americans expected to have at least one anxiety disorder in the course of their lives is staggering: one in three, ages 13 or older, according to researchers at Harvard Medical School and Technische Universität Dresden. The number is even higher for women—about 40%. Each year, about 40 million American adults have an anxiety disorder, and that doesn't include the millions of garden-variety worriers and insomniacs whose anxiety, though not debilitating, leaches away joy and steals their peace of mind.

Rates of anxiety disorders, as well as depression, seem to be increasing among young people, particularly college students. According to a 2016 survey by the American College Health Association, 17% of students were diagnosed with or treated for anxiety problems during the previous year, and nearly 14% were diagnosed with or treated for depression. That is up from about 10% each for anxiety and depression in 2008.

A certain amount of anxiety is a good thing, motivating us to study for tests, prepare for presentations, save for retirement. Too much anxiety, however, can be incapacitating and expensive. Anxiety disorders cost the U.S. about \$63 billion a year, according to a 1999 study published in the Journal of Clinical Psychiatry, the most recent estimate available. The tally includes doctor and hospital visits, psychiatric treatment, prescription drugs and the value of lost productivity at work. Anxiety can also lead to depression, substance abuse and even suicide. Anxious people who work have lower incomes. They are less likely to marry, and, if they do, more likely to divorce.

Scientists have found that many childhood events and experiences—from illness to trauma to overprotective and controlling styles of parenting—can contribute to the development of anxiety. It is also partly genetic. Having a first-degree relative—a parent, sibling or child—with an anxiety disorder bumps a person's risk of developing one by up to five times that of the general population.

I certainly don't see my anxiety as a gift, but it has some upsides. When I'm avoiding some necessary confrontation or saying yes to too many superfluous obligations, I feel it, and it kicks me into action. Weirdly, anxiety makes me live a more authentic life. And a more empathic one. It has made me feel vulnerable and more open to asking for help, thereby deepening my friendships.

People who have a brush with death often talk of how it has given them a sense of what really matters. An omnipresent fear of disaster and a constant bracing for catastrophe can do that, too. Time takes on more urgency. The background hum of uneasiness in my mind has motivated me to work harder, to speak more honestly and, curiously, to take more risks than I might otherwise have. Anxiety means that I'm simply not mellow enough to take things for granted. And that has made my life all the richer.

Q.11 [11594329]

Which of the following has been termed unfortunate by the author?

- 1 ☐ The non-discovery of the root cause of anxiety disorders.
- 2 ☐ The rapid progress scientists are making to untangle the mysteries of anxiety disorder.
- 3 ☐ An increase in the number of young people suffering from anxiety disorder and depression.
- 4 ☐ The overwhelming number of Americans who are vulnerable to anxiety disorders.



Solution:

Correct Answer : 1

Your Answer : 1

Refer to the lines “Unfortunately, even as we have learned more about the varieties of anxiety in recent decades, the underlying causes of the disorder have remained poorly understood.” 1 is the clear answer.

 **Answer key/Solution**

Bookmark

FeedBack

Direction for questions (10-14): Read the given passage and answer the questions that follow.

I didn't know it at the time, but I was suffering from severe anxiety, a medical condition recognized by doctors going as far back as the ancient Greeks. Today, 11 different anxiety disorders are listed in the "Diagnostic and Statistical Manual of Mental Disorders," the bible for mental-health professionals, and they are being diagnosed in rising numbers.

Unfortunately, even as we have learned more about the varieties of anxiety in recent decades, the underlying causes of the disorder have remained poorly understood. But that is changing. Scientists have started to unravel some of the mysteries of the anxious brain, and their discoveries may soon lead to new and better treatments.

Still, rates of anxiety disorders, as well as depression, seem to be increasing among young people. The number of Americans expected to have at least one anxiety disorder in the course of their lives is staggering: one in three, ages 13 or older, according to researchers at Harvard Medical School and Technische Universität Dresden. The number is even higher for women—about 40%. Each year, about 40 million American adults have an anxiety disorder, and that doesn't include the millions of garden-variety worriers and insomniacs whose anxiety, though not debilitating, leaches away joy and steals their peace of mind.

Rates of anxiety disorders, as well as depression, seem to be increasing among young people, particularly college students. According to a 2016 survey by the American College Health Association, 17% of students were diagnosed with or treated for anxiety problems during the previous year, and nearly 14% were diagnosed with or treated for depression. That is up from about 10% each for anxiety and depression in 2008.

A certain amount of anxiety is a good thing, motivating us to study for tests, prepare for presentations, save for retirement. Too much anxiety, however, can be incapacitating and expensive. Anxiety disorders cost the U.S. about \$63 billion a year, according to a 1999 study published in the Journal of Clinical Psychiatry, the most recent estimate available. The tally includes doctor and hospital visits, psychiatric treatment, prescription drugs and the value of lost productivity at work. Anxiety can also lead to depression, substance abuse and even suicide. Anxious people who work have lower incomes. They are less likely to marry, and, if they do, more likely to divorce.

Scientists have found that many childhood events and experiences—from illness to trauma to overprotective and controlling styles of parenting—can contribute to the development of anxiety. It is also partly genetic. Having a first-degree relative—a parent, sibling or child—with an anxiety disorder bumps a person's risk of developing one by up to five times that of the general population.

I certainly don't see my anxiety as a gift, but it has some upsides. When I'm avoiding some necessary confrontation or saying yes to too many superfluous obligations, I feel it, and it kicks me into action. Weirdly, anxiety makes me live a more authentic life. And a more empathic one. It has made me feel vulnerable and more open to asking for help, thereby deepening my friendships.

People who have a brush with death often talk of how it has given them a sense of what really matters. An omnipresent fear of disaster and a constant bracing for catastrophe can do that, too. Time takes on more urgency. The background hum of uneasiness in my mind has motivated me to work harder, to speak more honestly and, curiously, to take more risks than I might otherwise have. Anxiety means that I'm simply not mellow enough to take things for granted. And that has made my life all the richer.

Q.12 [11594329]

What is the fundamental conclusion that the author of the passage has made?

- 1 ☐ Despite his/her anxiety, the author has lived a more meaningful life.
- 2 ☐ Anxiety disorders have started to take the shape of an epidemic in the USA.
- 3 ☐ Scientists have made some insignificant progress in the field of treating anxiety disorders.
- 4 ☐ The search for a complete comprehension of anxiety disorders, an increasing risk for humans, is in the nascent stage.

Solution:

Correct Answer : 4

 Answer key/Solution

This can be answered by the method of elimination. 1 is misleading because the author mentions his/her anxiety as a reason for adding more value to his/her life.

2 is wrong because the word "epidemic" is too broad for this passage. 3 is wrong because of the word "insignificant". 4 is the best answer as the author clearly mentions this issue in the first three paragraphs.

Bookmark

FeedBack

Direction for questions (10-14): Read the given passage and answer the questions that follow.

I didn't know it at the time, but I was suffering from severe anxiety, a medical condition recognized by doctors going as far back as the ancient Greeks. Today, 11 different anxiety disorders are listed in the "Diagnostic and Statistical Manual of Mental Disorders," the bible for mental-health professionals, and they are being diagnosed in rising numbers.

Unfortunately, even as we have learned more about the varieties of anxiety in recent decades, the underlying causes of the disorder have remained poorly understood. But that is changing. Scientists have started to unravel some of the mysteries of the anxious brain, and their discoveries may soon lead to new and better treatments.

Still, rates of anxiety disorders, as well as depression, seem to be increasing among young people. The number of Americans expected to have at least one anxiety disorder in the course of their lives is staggering: one in three, ages 13 or older, according to researchers at Harvard Medical School and Technische Universität Dresden. The number is even higher for women—about 40%. Each year, about 40 million American adults have an anxiety disorder, and that doesn't include the millions of garden-variety worriers and insomniacs whose anxiety, though not debilitating, leaches away joy and steals their peace of mind.

Rates of anxiety disorders, as well as depression, seem to be increasing among young people, particularly college students. According to a 2016 survey by the American College Health Association, 17% of students were diagnosed with or treated for anxiety problems during the previous year, and nearly 14% were diagnosed with or treated for depression. That is up from about 10% each for anxiety and depression in 2008.

A certain amount of anxiety is a good thing, motivating us to study for tests, prepare for presentations, save for retirement. Too much anxiety, however, can be incapacitating and expensive. Anxiety disorders cost the U.S. about \$63 billion a year, according to a 1999 study published in the Journal of Clinical Psychiatry, the most recent estimate available. The tally includes doctor and hospital visits, psychiatric treatment, prescription drugs and the value of lost productivity at work. Anxiety can also lead to depression, substance abuse and even suicide. Anxious people who work have lower incomes. They are less likely to marry, and, if they do, more likely to divorce.

Scientists have found that many childhood events and experiences—from illness to trauma to overprotective and controlling styles of parenting—can contribute to the development of anxiety. It is also partly genetic. Having a first-degree relative—a parent, sibling or child—with an anxiety disorder bumps a person's risk of developing one by up to five times that of the general population.

I certainly don't see my anxiety as a gift, but it has some upsides. When I'm avoiding some necessary confrontation or saying yes to too many superfluous obligations, I feel it, and it kicks me into action. Weirdly, anxiety makes me live a more authentic life. And a more empathic one. It has made me feel vulnerable and more open to asking for help, thereby deepening my friendships.

People who have a brush with death often talk of how it has given them a sense of what really matters. An omnipresent fear of disaster and a constant bracing for catastrophe can do that, too. Time takes on more urgency. The background hum of uneasiness in my mind has motivated me to work harder, to speak more honestly and, curiously, to take more risks than I might otherwise have. Anxiety means that I'm simply not mellow enough to take things for granted. And that has made my life all the richer.

Q.13 [11594329]

What can be inferred about anxiety disorder?

- 1 ☐ Though not debilitating, depression leaches away joy and steals the peace of mind of its victims.
- 2 ☐ Anxiety disorder can result in one's emotional as well as financial ruin.
- 3 ☐ Anxiety disorder leads to a more fulfilling life for the person suffering from it.
- 4 ☐ Women are more likely to suffer from anxiety disorders as compared to men.



Solution:

Correct Answer : 2

Your Answer : 2

1 is wrong because the passage talks about insomnia and other garden variety of anxiety disorder (not depression) while talking about "leaches away joy and ...". 2 can be inferred from the fifth paragraph. 3 is wrong because it doesn't match the tone of the author in the passage. 4 is wrong because the passage talks about American women, not women in general.

Bookmark

FeedBack

 Answer key/Solution

Direction for questions (10-14): Read the given passage and answer the questions that follow.

I didn't know it at the time, but I was suffering from severe anxiety, a medical condition recognized by doctors going as far back as the ancient Greeks. Today, 11 different anxiety disorders are listed in the "Diagnostic and Statistical Manual of Mental Disorders," the bible for mental-health professionals, and they are being diagnosed in rising numbers.

Unfortunately, even as we have learned more about the varieties of anxiety in recent decades, the underlying causes of the disorder have remained poorly understood. But that is changing. Scientists have started to unravel some of the mysteries of the anxious brain, and their discoveries may soon lead to new and better treatments.

Still, rates of anxiety disorders, as well as depression, seem to be increasing among young people. The number of Americans expected to have at least one anxiety disorder in the course of their lives is staggering: one in three, ages 13 or older, according to researchers at Harvard Medical School and Technische Universität Dresden. The number is even higher for women—about 40%. Each year, about 40 million American adults have an anxiety disorder, and that doesn't include the millions of garden-variety worriers and insomniacs whose anxiety, though not debilitating, leaches away joy and steals their peace of mind.

Rates of anxiety disorders, as well as depression, seem to be increasing among young people, particularly college students. According to a 2016 survey by the American College Health Association, 17% of students were diagnosed with or treated for anxiety problems during the previous year, and nearly 14% were diagnosed with or treated for depression. That is up from about 10% each for anxiety and depression in 2008.

A certain amount of anxiety is a good thing, motivating us to study for tests, prepare for presentations, save for retirement. Too much anxiety, however, can be incapacitating and expensive. Anxiety disorders cost the U.S. about \$63 billion a year, according to a 1999 study published in the Journal of Clinical Psychiatry, the most recent estimate available. The tally includes doctor and hospital visits, psychiatric treatment, prescription drugs and the value of lost productivity at work. Anxiety can also lead to depression, substance abuse and even suicide. Anxious people who work have lower incomes. They are less likely to marry, and, if they do, more likely to divorce.

Scientists have found that many childhood events and experiences—from illness to trauma to overprotective and controlling styles of parenting—can contribute to the development of anxiety. It is also partly genetic. Having a first-degree relative—a parent, sibling or child—with an anxiety disorder bumps a person's risk of developing one by up to five times that of the general population.

I certainly don't see my anxiety as a gift, but it has some upsides. When I'm avoiding some necessary confrontation or saying yes to too many superfluous obligations, I feel it, and it kicks me into action. Weirdly, anxiety makes me live a more authentic life. And a more empathic one. It has made me feel vulnerable and more open to asking for help, thereby deepening my friendships.

People who have a brush with death often talk of how it has given them a sense of what really matters. An omnipresent fear of disaster and a constant bracing for catastrophe can do that, too. Time takes on more urgency. The background hum of uneasiness in my mind has motivated me to work harder, to speak more honestly and, curiously, to take more risks than I might otherwise have. Anxiety means that I'm simply not mellow enough to take things for granted. And that has made my life all the richer.

Q.14 [11594329]

According to the passage, all of the following can contribute to the development of anxiety disorder except:

- 1 ☐ Hereditary factors
- 2 ☐ Childhood trauma
- 3 ☐ Overprotective parents
- 4 ☐ Constant fear of catastrophe



Solution:

Correct Answer : 4

Your Answer : 4

1, 2, and 3 are mentioned in the sixth paragraph as causes contributing to the development of anxiety disorders. 4 is mentioned in a different context in the last paragraph. Hence, 4 is the answer.

[🔍 Answer key/Solution](#)

Bookmark

FeedBack

Direction for questions (15-18): Read the given passage and answer the questions that follow.

Historically, varying definitions of the notion of the 'nation' have endured. Two key events heralded the advent of the modern nation-state; the American (1776) and French (1789) Revolutions. Both saw the emergence of beliefs in the sovereignty of the people united under a legal system underpinned by a written constitution. Although issues of social inequalities and cultural diversity complicated the political-legal model in both the American and French cases, both nations used notions of linguistic uniformity to cement the nation.

Since the end of the Cold War a more complex world has emerged, characterized by increasing diversity, related reassertions of national identities and the construction of new forms of state control. In the context of democracy, understandings of citizenships are changing as people come to view civic engagement and participation as more effective means of achieving social change than traditional forms of political representation. Political institutions are having to adjust in order to accommodate hitherto marginalized communities more effectively within democratic processes. As nations increasingly come under the influence of global population flows and transnational networks, the issue of cultural diversity becomes ever more crucial.

Western conceptions of secularism are shaped by a context of predominantly single-religion societies. In recent times, however, this conception of secularism and the strategies that flow from it have increasingly come under severe strain. This is largely due to intensified globalization and the migration of millions of people into the metropolis from former colonies. One consequence has been to bring to the West a form of deep religious diversity. It is in this context that the model of secularism developed in the sub-continent, especially in India, provides an alternative conception that offers lessons for how the state can respond to deep religious diversity.

When India became independent it was deeply conscious of the two types of domination; intra- and inter-religious domination. This created the need to ensure that whatever secularism was employed in India met the needs of a deeply religiously diverse society while also complying with the principles of freedom and equality. Hence, within the Indian model there was never a time when people felt that religious should be banished from the public sphere: and this fact was critical to the manner in which the Indian Constitution was shaped. In responding to the complex and religiously diverse context of India, the position of non-establishment was taken in order to ensure that all citizens could identify with the state.

The Indian model adopts the concept of principled distance. This concept entails a flexible approach to the question of inclusion/exclusion of religion and the engagement/ disengagement of the state. It allows, for example, that a practice that is banned or regulated in one culture can be permitted in the minority culture because of the distinctive status and meaning it has for its members.

The Indian approach enables value-based pragmatism as the boundaries between state and religion are porous. The state can intervene in religions, to help or hinder them. This is reflected in the multiple roles the state may take in relating to religion. The practice of contextual secularism requires a model of moral reasoning that recognizes that the conflict between individual and group rights cannot always be adjudicated by recourse to general or abstract principles. The Indian model accommodates the fact that there are many values and instead of choosing one or the other, encourages you to choose both. This commitment to multiple values, principled distance and contextual secularism means that the state tries to balance different, ambiguous but equally important values.

Q.15 [11594329]

Which of the following can be identified as a possible central theme of the passage?

-
- 1 ☐ The evolving notions of nations and citizenship and the impact of globalization on them.
-
- 2 ☐ The enduring impact of American and French revolutions on the socio-political fabric of world societies.
-
- 3 ☐ The evolving ideas of nations and secularism and the relevance of the Indian model of secularism.
-
- 4 ☐ India's model of secularism, its key features and its impact on the world at large.
-

Solution:

Correct Answer : 3

 Answer key/Solution

The passage talks about the changing notions of nations, citizenship and secularism and then goes to discuss the relevance of India's model of secularism.

Thus, option 3 captures the essence of the passage holistically. The other options are either restricted or go beyond the scope of the passage.

Bookmark

FeedBack

Direction for questions (15-18): Read the given passage and answer the questions that follow.

Historically, varying definitions of the notion of the 'nation' have endured. Two key events heralded the advent of the modern nation-state; the American (1776) and French (1789) Revolutions. Both saw the emergence of beliefs in the sovereignty of the people united under a legal system underpinned by a written constitution. Although issues of social inequalities and cultural diversity complicated the political-legal model in both the American and French cases, both nations used notions of linguistic uniformity to cement the nation.

Since the end of the Cold War a more complex world has emerged, characterized by increasing diversity, related reassertions of national identities and the construction of new forms of state control. In the context of democracy, understandings of citizenships are changing as people come to view civic engagement and participation as more effective means of achieving social change than traditional forms of political representation. Political institutions are having to adjust in order to accommodate hitherto marginalized communities more effectively within democratic processes. As nations increasingly come under the influence of global population flows and transnational networks, the issue of cultural diversity becomes ever more crucial.

Western conceptions of secularism are shaped by a context of predominantly single-religion societies. In recent times, however, this conception of secularism and the strategies that flow from it have increasingly come under severe strain. This is largely due to intensified globalization and the migration of millions of people into the metropolis from former colonies. One consequence has been to bring to the West a form of deep religious diversity. It is in this context that the model of secularism developed in the sub-continent, especially in India, provides an alternative conception that offers lessons for how the state can respond to deep religious diversity.

When India became independent it was deeply conscious of the two types of domination; intra- and inter-religious domination. This created the need to ensure that whatever secularism was employed in India met the needs of a deeply religiously diverse society while also complying with the principles of freedom and equality. Hence, within the Indian model there was never a time when people felt that religious should be banished from the public sphere: and this fact was critical to the manner in which the Indian Constitution was shaped. In responding to the complex and religiously diverse context of India, the position of non-establishment was taken in order to ensure that all citizens could identify with the state.

The Indian model adopts the concept of principled distance. This concept entails a flexible approach to the question of inclusion/exclusion of religion and the engagement/ disengagement of the state. It allows, for example, that a practice that is banned or regulated in one culture can be permitted in the minority culture because of the distinctive status and meaning it has for its members.

The Indian approach enables value-based pragmatism as the boundaries between state and religion are porous. The state can intervene in religions, to help or hinder them. This is reflected in the multiple roles the state may take in relating to religion. The practice of contextual secularism requires a model of moral reasoning that recognizes that the conflict between individual and group rights cannot always be adjudicated by recourse to general or abstract principles. The Indian model accommodates the fact that there are many values and instead of choosing one or the other, encourages you to choose both. This commitment to multiple values, principled distance and contextual secularism means that the state tries to balance different, ambiguous but equally important values.

Q.16 [11594329]

Which of the following could be a possible logical corollary to the passage?

- 1 ☐ This in turn makes it possible bring together seemingly incompatible values - a great and important strength of Indian secularism.
- 2 ☐ This raises grave concerns about India's attempt to bring together seemingly incompatible values - an impossible task.
- 3 ☐ The conflict between Western conceptions of secularism and India's religious realities have forced India to take this hitherto unexplored road.
- 4 ☐ The changing notions of state control have led the various world governments to re-evaluate the extent to which the state should drive the social discourse.

Solution:**Correct Answer : 1**[🔍 Answer key/Solution](#)

The last paragraph of the passage talks about the advantages of India's model of secularism and why it has worked for the country. Thus, the next line of the paragraph will further build upon this idea. Thus, option 1 is the correct answer. Option 2 changes the tone adopted by the author while the other two options modify the idea being discussed.

[Bookmark](#)[FeedBack](#)

Direction for questions (15-18): Read the given passage and answer the questions that follow.

Historically, varying definitions of the notion of the 'nation' have endured. Two key events heralded the advent of the modern nation-state; the American (1776) and French (1789) Revolutions. Both saw the emergence of beliefs in the sovereignty of the people united under a legal system underpinned by a written constitution. Although issues of social inequalities and cultural diversity complicated the political-legal model in both the American and French cases, both nations used notions of linguistic uniformity to cement the nation.

Since the end of the Cold War a more complex world has emerged, characterized by increasing diversity, related reassertions of national identities and the construction of new forms of state control. In the context of democracy, understandings of citizenships are changing as people come to view civic engagement and participation as more effective means of achieving social change than traditional forms of political representation. Political institutions are having to adjust in order to accommodate hitherto marginalized communities more effectively within democratic processes. As nations increasingly come under the influence of global population flows and transnational networks, the issue of cultural diversity becomes ever more crucial.

Western conceptions of secularism are shaped by a context of predominantly single-religion societies. In recent times, however, this conception of secularism and the strategies that flow from it have increasingly come under severe strain. This is largely due to intensified globalization and the migration of millions of people into the metropolis from former colonies. One consequence has been to bring to the West a form of deep religious diversity. It is in this context that the model of secularism developed in the sub-continent, especially in India, provides an alternative conception that offers lessons for how the state can respond to deep religious diversity.

When India became independent it was deeply conscious of the two types of domination; intra- and inter-religious domination. This created the need to ensure that whatever secularism was employed in India met the needs of a deeply religiously diverse society while also complying with the principles of freedom and equality. Hence, within the Indian model there was never a time when people felt that religious should be banished from the public sphere: and this fact was critical to the manner in which the Indian Constitution was shaped. In responding to the complex and religiously diverse context of India, the position of non-establishment was taken in order to ensure that all citizens could identify with the state.

The Indian model adopts the concept of principled distance. This concept entails a flexible approach to the question of inclusion/exclusion of religion and the engagement/ disengagement of the state. It allows, for example, that a practice that is banned or regulated in one culture can be permitted in the minority culture because of the distinctive status and meaning it has for its members.

The Indian approach enables value-based pragmatism as the boundaries between state and religion are porous. The state can intervene in religions, to help or hinder them. This is reflected in the multiple roles the state may take in relating to religion. The practice of contextual secularism requires a model of moral reasoning that recognizes that the conflict between individual and group rights cannot always be adjudicated by recourse to general or abstract principles. The Indian model accommodates the fact that there are many values and instead of choosing one or the other, encourages you to choose both. This commitment to multiple values, principled distance and contextual secularism means that the state tries to balance different, ambiguous but equally important values.

Q.17 [11594329]

Based on the first two paragraphs, all the following statements regarding nation states are true EXCEPT:

- 1 ☐ Both America and France applied notions of linguistic uniformity to reinforce the idea of national identity.
-
- 2 ☐ World over, the traditional forms of political representation have been rejected in favor of civic engagement and participation.
-
- 3 ☐ The issue of cultural diversity is gaining growing importance owing to global population flows and transnational networks.
-
- 4 ☐ The notions of national identities and state control are being reasserted and reestablished globally to reflect the emerging world order.
-

Solution:

Correct Answer : 2

 Answer key/Solution

All the other statements can be clearly inferred from the first two paragraphs of the passage. The second option distorts the intended meaning of the author. The paragraph simply states that *'In the context of democracy, understandings of citizenships are changing as people come to view civic engagement and participation as more effective means of achieving social change than traditional forms of political representation.'*

Bookmark

FeedBack

Direction for questions (15-18): Read the given passage and answer the questions that follow.

Historically, varying definitions of the notion of the 'nation' have endured. Two key events heralded the advent of the modern nation-state; the American (1776) and French (1789) Revolutions. Both saw the emergence of beliefs in the sovereignty of the people united under a legal system underpinned by a written constitution. Although issues of social inequalities and cultural diversity complicated the political-legal model in both the American and French cases, both nations used notions of linguistic uniformity to cement the nation.

Since the end of the Cold War a more complex world has emerged, characterized by increasing diversity, related reassertions of national identities and the construction of new forms of state control. In the context of democracy, understandings of citizenships are changing as people come to view civic engagement and participation as more effective means of achieving social change than traditional forms of political representation. Political institutions are having to adjust in order to accommodate hitherto marginalized communities more effectively within democratic processes. As nations increasingly come under the influence of global population flows and transnational networks, the issue of cultural diversity becomes ever more crucial.

Western conceptions of secularism are shaped by a context of predominantly single-religion societies. In recent times, however, this conception of secularism and the strategies that flow from it have increasingly come under severe strain. This is largely due to intensified globalization and the migration of millions of people into the metropolis from former colonies. One consequence has been to bring to the West a form of deep religious diversity. It is in this context that the model of secularism developed in the sub-continent, especially in India, provides an alternative conception that offers lessons for how the state can respond to deep religious diversity.

When India became independent it was deeply conscious of the two types of domination; intra- and inter-religious domination. This created the need to ensure that whatever secularism was employed in India met the needs of a deeply religiously diverse society while also complying with the principles of freedom and equality. Hence, within the Indian model there was never a time when people felt that religious should be banished from the public sphere: and this fact was critical to the manner in which the Indian Constitution was shaped. In responding to the complex and religiously diverse context of India, the position of non-establishment was taken in order to ensure that all citizens could identify with the state.

The Indian model adopts the concept of principled distance. This concept entails a flexible approach to the question of inclusion/exclusion of religion and the engagement/ disengagement of the state. It allows, for example, that a practice that is banned or regulated in one culture can be permitted in the minority culture because of the distinctive status and meaning it has for its members.

The Indian approach enables value-based pragmatism as the boundaries between state and religion are porous. The state can intervene in religions, to help or hinder them. This is reflected in the multiple roles the state may take in relating to religion. The practice of contextual secularism requires a model of moral reasoning that recognizes that the conflict between individual and group rights cannot always be adjudicated by recourse to general or abstract principles. The Indian model accommodates the fact that there are many values and instead of choosing one or the other, encourages you to choose both. This commitment to multiple values, principled distance and contextual secularism means that the state tries to balance different, ambiguous but equally important values.

Q.18 [11594329]

All of the following are true with respect to the Indian model of secularism EXCEPT:

- 1 ☐ During the shaping of the constitution, it was realized that religion could in no way be allowed to be a part of the political sphere.
- 2 ☐ In the Indian approach to secularism, the boundaries between state and religion are porous.
- 3 ☐ Contextual secularism recognizes that the conflict between individual and group rights cannot always be adjudicated by recourse to general or abstract principles.
- 4 ☐ The state tries to balance different, ambiguous but equally important values through commitment to multiple values, principled distance and contextual secularism.

Solution:

Correct Answer : 1

 Answer key/Solution

All the statements can be directly identified from the concluding paragraphs except the first option. Refer the lines, 'Hence, within the Indian model there was never a time when people felt that religious should be banished from the public sphere: and this fact was critical to the manner in which the Indian Constitution was shaped.'

Bookmark

FeedBack

Q.19 [11594329]

Directions for question (19): The four sentences (labelled 1, 2, 3, and 4) given in this question, when properly sequenced, form a coherent paragraph. Decide on the proper order for the sentences and key in this sequence of four numbers as your answer.

1. We can't get by without them, so they end up being justified by the pragmatic standards tailored to supplant them.
2. The worry here is that we have a practical, emotional need for comprehensive worldviews.
3. The more serious threat of self-refutation is contained in a quip "Sidney used to say that pragmatism was certainly true; the only problem was that it didn't work."
4. The motivation to construct a worldview, then, is not epistemological.



Solution:

Correct Answer : 3214

Your Answer : 3214

 Answer key/Solution

Threat and worry are related ideas. 2 and 1 form mandatory pairs as the 'them' in 1 refers to worldviews in 2. 4 gives a conclusion and the word 'then' shows that it should come after 2 and 1. So 214 becomes a sequence. 3 is related to 2, so the sequence becomes 3214.

Bookmark

FeedBack

Q.20 [11594329]

Directions for question (20): The four sentences (labelled 1, 2, 3, and 4) given in this question, when properly sequenced, form a coherent paragraph. Decide on the proper order for the sentences and key in this sequence of four numbers as your answer.

1. War may have other advantages, but the cost is always too great.
2. It facilitates war and by facilitating wars, which are mostly unnecessary and unjustifiable according to historical record, increases the total amount of human suffering.
3. I say patriotism is a mistake because for patriots, it is a moral mistake.
4. It is almost always the direct or indirect source of vice, harm, the infliction and endurance of suffering.



Solution:

Correct Answer : 3421

Your Answer : 3421

3 is the topic sentence. 4 gives the drawbacks of patriotism. 2 relates the suffering to wars. 1 talks about the cost of war. Hence, 3421.

Bookmark

FeedBack

Answer key/Solution

Q.21 [11594329]

Directions for question (21): Five sentences related to a topic are given below. Four of them can be put together to form a meaningful and coherent short paragraph. Identify the odd one out.

1. In Sweden, birds' eggs were often placed in cremated ashes.
2. It is hard to come to any overall conclusions about Viking belief from their burial practices.
3. Burying someone with two horses could be a kind of tribute, but why bisect the horses and then swap the halves over?
4. Archaeologists avoid such speculation, though we have no good explanations for this.
5. They performed cremation and inhumation, boat burial and barrow burial.



Solution:

Correct Answer : 4

Your Answer : 3

4 is the odd sentence out. No speculation has been made in the paragraph. The correct sequence is 2513.

Bookmark

FeedBack

Answer key/Solution

Q.22 [11594329]

Directions for question (22): The passage given below is followed by four summaries. Choose the option that best captures the author's position.

Both Western and Indian philosophical traditions judge that ordinary anger is retributive; what I have called transition anger is exceptional. Studying the breakdown of marriages and friendships, one is inclined to agree. However, the numbers don't matter; it is the distinction that matters, and this distinction has simply not been clearly made, throughout the whole philosophical tradition. Transition anger is useful in a struggle and does not burden the personality. Retributive anger burdens the personality—and is not very useful in a struggle for freedom. Martin Luther King, Jr., the one distinguished Western philosopher who did recognize and emphasize this distinction, spoke of the way that the anger of people in his movement had to be purified and "channelized."

- 1 ☐ There are two types of anger--retributive that burdens the personality and is not useful in a struggle for freedom, and transition that does not do either of these.
- 2 ☐ Both western and Indian philosophical traditions have not clearly made the distinction between retributive anger and transition anger clearly.
- 3 ☐ The distinction between ordinary, retributive anger and transition anger in burdening the personality and usefulness in a struggle for freedom must be recognised.
- 4 ☐ The distinction between retributive anger, which burdens the personality and is not useful in a struggle, and transition anger which does not do these is important.

Solution:

Correct Answer : 4

 Answer key/Solution

There are not two given types of anger. It is the author who calls a type of anger as transition anger. So, 1 is eliminated. 2 is not the main point of the para.

Between 3 and 4, 3 goes a little further in saying that this distinction 'must be recognised'. The author simply states that this distinction matters, and Martin Luther King Jr. recognised it.

Bookmark

FeedBack

Q.23 [11594329]

Directions for question (23): Five sentences related to a topic are given below. Four of them can be put together to form a meaningful and coherent short paragraph. Identify the odd one out.

1. Modern technologies have visibly increased opportunities for random shaming.
2. Several explanations may combine as to why the problem of shaming is so pervasive.
3. Some people are simply nasty and enjoy the opportunity to inflict harm on others.
4. Then there are conservatives who simply do not buy into the fashionable dismissal of shame.
5. An individual tweets a comment giving an unconventional take on a controversial issue, and counter-tweets swarm, some simply delighted at the opportunity to cause pain.



Solution:**Correct Answer : 4****Your Answer : 4**

2 is the introductory sentence. 3 gives an explanation and has to follow 2. 1 and 5 make a mandatory pair as 5 is an example of 1. The sequence is 2315. 4 is the odd sentence.

Bookmark

FeedBack

 Answer key/Solution**Q.24 [11594329]**

Directions for question (24): The four sentences (labelled 1, 2, 3, and 4) given in this question, when properly sequenced, form a coherent paragraph. Decide on the proper order for the sentences and key in this sequence of four numbers as your answer.

1. On 10 July 1940, exactly 80 years ago, the French parliament passed an enabling act granting full power to Marshal Philippe Pétain.
2. Thus, dynastic politicians may contribute to stabilising democracies by better resisting peer pressure.
3. Rising populism has raised concerns that democracies may give in to authoritarian pressure.
4. Analysing how the Members of Parliament voted, this column shows that MPs belonging to a pro-democratic dynasty were more likely to oppose the act.

✕

Solution:**Correct Answer : 3142****Your Answer : 3412**

142 forms a sequence as 1 talks about an act and 4 furthers that discussion, and 4 talks about dynastic politicians and 2 furthers that discussion. 3 is the topic sentence, the present scenario, in the context of which 1,4, and 2 are stated. 3142.

Bookmark

FeedBack

 Answer key/Solution**Q.25 [11594329]**

Directions for question (25): Five sentences related to a topic are given below. Four of them can be put together to form a meaningful and coherent short paragraph. Identify the odd one out.

1. But when you're writing about an experience as timeless as love, it's pretty hard to step outside the canon and offer a new reading of the subject.
2. That sort of approach doesn't appeal to me.
3. The more so as the people I study are not imagined literary characters but my next-door neighbours.
4. I've been thinking about these two books for a long time.
5. Gathering the shards of truth isn't easy.

**Solution:****Correct Answer : 2****Your Answer : 4**[🔍 Answer key/Solution](#)

4 and 1 form a pair. Similarly, 1 and 3 form a pair because it further explains why it is hard for the author. 5 is related to this paragraph because we talk about a person writing not about imagined characters but about people in real life who live next-door to her. The sequence is 5413. 2 is the odd sentence out because no particular approach is mentioned in the para, and the author talks about her difficulty in writing, not her likes and dislikes in writing.

[Bookmark](#)[FeedBack](#)**Q.26 [11594329]**

Directions for question (26): The passage given below is followed by four summaries. Choose the option that best captures the author's position.

One myth that won't seem to go away is that DTP vaccine causes sudden infant death syndrome (SIDS). This belief came about because a moderate proportion of children who die of SIDS have recently been vaccinated with DTP; on the surface, this seems to point toward a causal connection. This logic is faulty however; you might as well say that eating bread causes car crashes, since most drivers who crash their cars could probably be shown to have eaten bread within the past 24 hours. If you consider that most SIDS deaths occur during the age range when three shots of DTP are given, you would expect DTP shots to precede a fair number of SIDS deaths simply by chance. In fact, when a number of well-controlled studies were conducted during the 1980s, the investigators found, nearly unanimously, that the number of SIDS deaths temporally associated with DTP vaccination was within the range expected to occur by chance. In other words, the SIDS deaths would have occurred even if no vaccinations had been given.

- 1 ☐ Blaming DTP vaccine for SIDS is a logical fallacy; it is also analogous to blaming eating bread for car crashes.
- 2 ☐ A number of studies have proven that DTP vaccines are not responsible for SIDS alone.
- 3 ☐ The myth that DTP vaccine causes SIDS is deeply entrenched in the minds of people and even proven research works are unable to eradicate this myth.
- 4 ☐ Studies conducted during the 1980s concluded that blaming DTP vaccines for SIDS is logically fallacious; yet, the myth seems to persist.



Solution:**Correct Answer : 4****Your Answer : 4**[Answer key/Solution](#)

The paragraph talks about the myth that DTP vaccine causes SIDS. It also cites the research works done in the 1980s. 1 is partially correct. 2 is a misleading option as it seems to suggest that DTP vaccines are responsible for many other things too. 3 is wrong because the phrase “deeply entrenched in the minds of people” is an extreme conclusion which is not mentioned in the paragraph. 4 is the best answer as it talks about the two main points discussed in the paragraph.

[Bookmark](#)[FeedBack](#)

Sec 2

Directions for questions 27 to 30: Answer the questions on the basis of the information given below.

Robin started her train journey from New Delhi to Kolkata. There are 9 stoppages – A, B, C, D, E, F, G, H, and K (not necessarily in this order) in between New Delhi and Kolkata. Robin at the start of the journey has downloaded a mobile application which enabled her to calculate the distance travelled by her. Based on Robin's observation, the following table is given which gives the distance (in km) between two stoppages at a time.

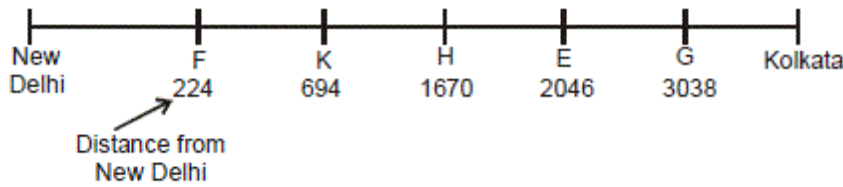
	E	F	G	H	K
New Delhi	2046	224	3038	1670	694
A	474	1348	1566	98	878
B	572	2394	420	948	1924
C	1534	288	2526	1158	182
D	802	1020	1794	426	550

Q.27 [11594329]

Find the distance (in km) of the 5th stoppage from New Delhi.

Solution:**Correct Answer : 1572**[Answer key/Solution](#)

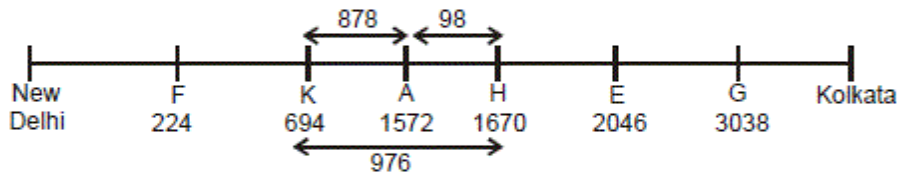
As we know these 9 stoppages are in between New Delhi and Kolkata, and from the first row of the table we know the positions of stoppages E, F, G, H and K from New Delhi. So, same can be placed as below–



[Note: Values written below the alphabets are the distances (in km) from New Delhi.]

Now we have to place A, B, C and D based on data given in table.

In the second row, A's distance from H and K is 98 and 878 which adds upto 976 and in the above made case distance between H and K is 976. So, A is in between K and H.

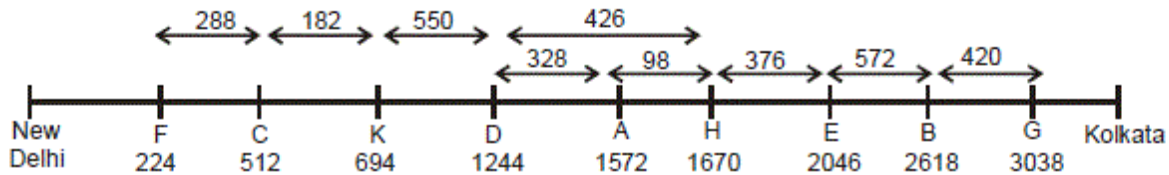


From 3rd row, we see the distances of B from E and G are 572 and 420 which adds upto 992 and in the above figure, the distance between E and G is 992, so B is in between E and G.

Similarly, from 4th row, the distances of C from F and K are 288 and 182 which adds upto 470 which is the distance between F and K, so C is in between F and K.

And also from row 5th, D is at distance 426 and 550 from H and K respectively, and the distance between H and K is 976 so D is in between H and K and we can also conclude it will be in between K and A.

So, finally we get,



A is the 5th stoppage. So, its distance from New Delhi is 1572 km.

[Bookmark](#)[Feedback](#)

Directions for questions 27 to 30: Answer the questions on the basis of the information given below.

Robin started her train journey from New Delhi to Kolkata. There are 9 stoppages – A, B, C, D, E, F, G, H, and K (not necessarily in this order) in between New Delhi and Kolkata. Robin at the start of the journey has downloaded a mobile application which enabled her to calculate the distance travelled by her. Based on Robin's observation, the following table is given which gives the distance (in km) between two stoppages at a time.

	E	F	G	H	K
New Delhi	2046	224	3038	1670	694
A	474	1348	1566	98	878
B	572	2394	420	948	1924
C	1534	288	2526	1158	182
D	802	1020	1794	426	550

Q.28 [11594329]

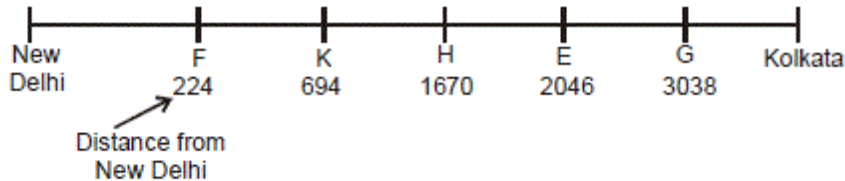
What is the absolute difference (in km) between the distance between stoppages B and G and that between C and H?

Solution:

Correct Answer : 738

[Answer key/Solution](#)

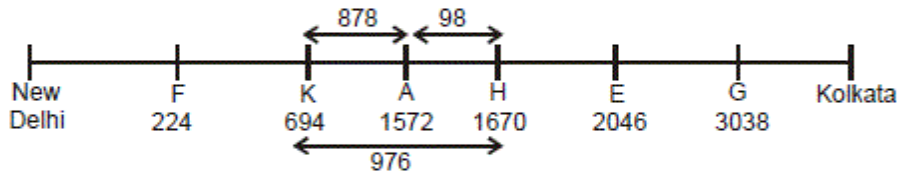
As we know these 9 stoppages are in between New Delhi and Kolkata, and from the first row of the table we know the positions of stoppages E, F, G, H and K from New Delhi. So, same can be placed as below–



[Note: Values written below the alphabets are the distances (in km) from New Delhi.]

Now we have to place A, B, C and D based on data given in table.

In the second row, A's distance from H and K is 98 and 878 which adds upto 976 and in the above made case distance between H and K is 976. So, A is in between K and H.

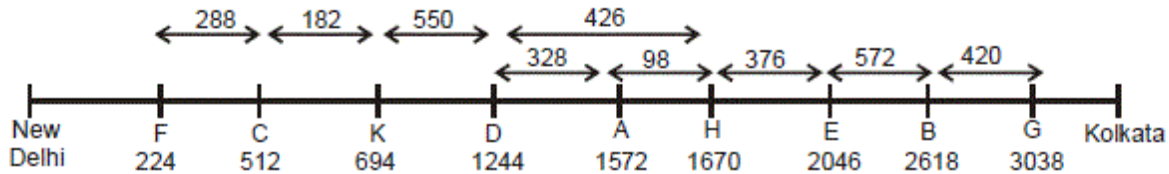


From 3rd row, we see the distances of B from E and G are 572 and 420 which adds upto 992 and in the above figure, the distance between E and G is 992, so B is in between E and G.

Similarly, from 4th row, the distances of C from F and K are 288 and 182 which adds upto 470 which is the distance between F and K, so C is in between F and K.

And also from row 5th, D is at distance 426 and 550 from H and K respectively, and the distance between H and K is 976 so D is in between H and K and we can also conclude it will be in between K and A.

So, finally we get,



The absolute difference between the distance between stoppages B and G and that between C and H is $|420 - 1158| = 738$ km.

[Bookmark](#)

[FeedBack](#)

Directions for questions 27 to 30: Answer the questions on the basis of the information given below.

Robin started her train journey from New Delhi to Kolkata. There are 9 stoppages – A, B, C, D, E, F, G, H, and K (not necessarily in this order) in between New Delhi and Kolkata. Robin at the start of the journey has downloaded a mobile application which enabled her to calculate the distance travelled by her. Based on Robin's observation, the following table is given which gives the distance (in km) between two stoppages at a time.

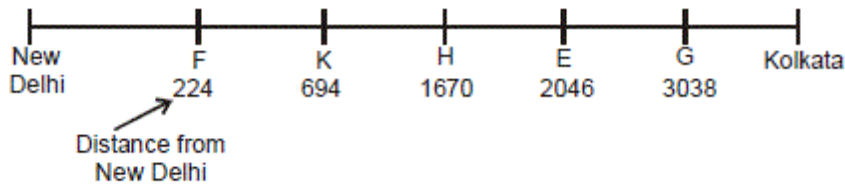
	E	F	G	H	K
New Delhi	2046	224	3038	1670	694
A	474	1348	1566	98	878
B	572	2394	420	948	1924
C	1534	288	2526	1158	182
D	802	1020	1794	426	550

Q.29 [11594329]

How many stoppages are there between C and B?

Solution:**Correct Answer : 5**[Answer key/Solution](#)

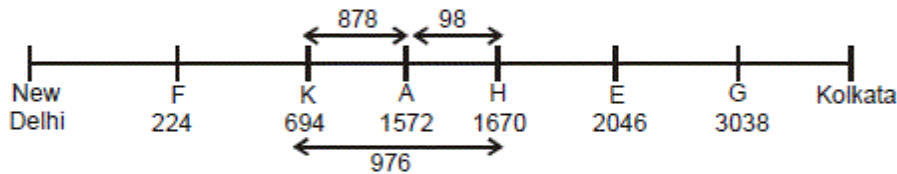
As we know these 9 stoppages are in between New Delhi and Kolkata, and from the first row of the table we know the positions of stoppages E, F, G, H and K from New Delhi. So, same can be placed as below–



[Note: Values written below the alphabets are the distances (in km) from New Delhi.]

Now we have to place A, B, C and D based on data given in table.

In the second row, A's distance from H and K is 98 and 878 which adds upto 976 and in the above made case distance between H and K is 976. So, A is in between K and H.

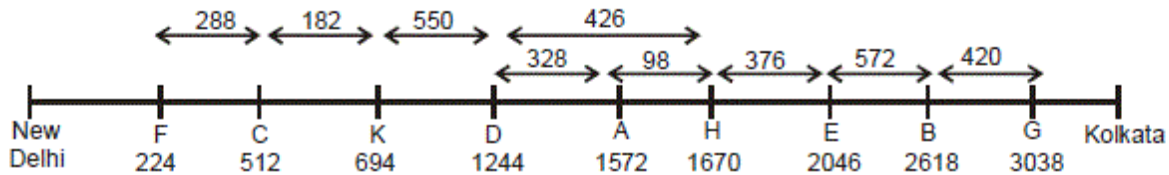


From 3rd row, we see the distances of B from E and G are 572 and 420 which adds upto 992 and in the above figure, the distance between E and G is 992, so B is in between E and G.

Similarly, from 4th row, the distances of C from F and K are 288 and 182 which adds upto 470 which is the distance between F and K, so C is in between F and K.

And also from row 5th, D is at distance 426 and 550 from H and K respectively, and the distance between H and K is 976 so D is in between H and K and we can also conclude it will be in between K and A.

So, finally we get,



Total 5 stoppages are there between C and B.

[Bookmark](#)
[FeedBack](#)

Directions for questions 27 to 30: Answer the questions on the basis of the information given below.

Robin started her train journey from New Delhi to Kolkata. There are 9 stoppages – A, B, C, D, E, F, G, H, and K (not necessarily in this order) in between New Delhi and Kolkata. Robin at the start of the journey has downloaded a mobile application which enabled her to calculate the distance travelled by her. Based on Robin's observation, the following table is given which gives the distance (in km) between two stoppages at a time.

	E	F	G	H	K
New Delhi	2046	224	3038	1670	694
A	474	1348	1566	98	878
B	572	2394	420	948	1924
C	1534	288	2526	1158	182
D	802	1020	1794	426	550

Q.30 [11594329]

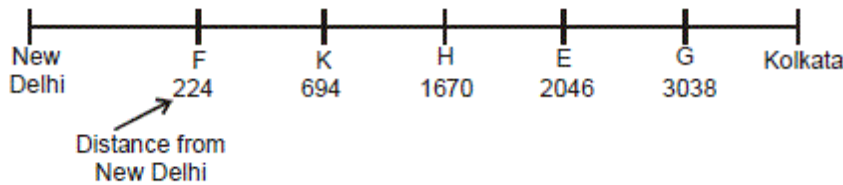
What is the distance (in km) between New Delhi and Kolkata if Kolkata is 568 km away from stoppage B?

Solution:

Correct Answer : 3186

[Answer key/Solution](#)

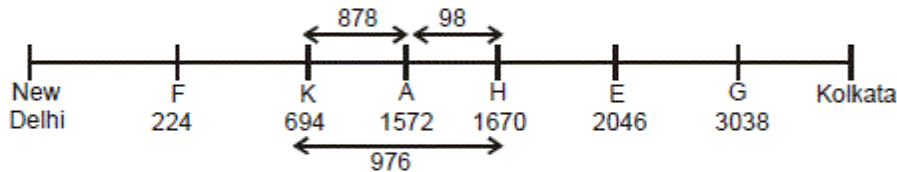
As we know these 9 stoppages are in between New Delhi and Kolkata, and from the first row of the table we know the positions of stoppages E, F, G, H and K from New Delhi. So, same can be placed as below—



[Note: Values written below the alphabets are the distances (in km) from New Delhi.]

Now we have to place A, B, C and D based on data given in table.

In the second row, A's distance from H and K is 98 and 878 which adds upto 976 and in the above made case distance between H and K is 976. So, A is in between K and H.

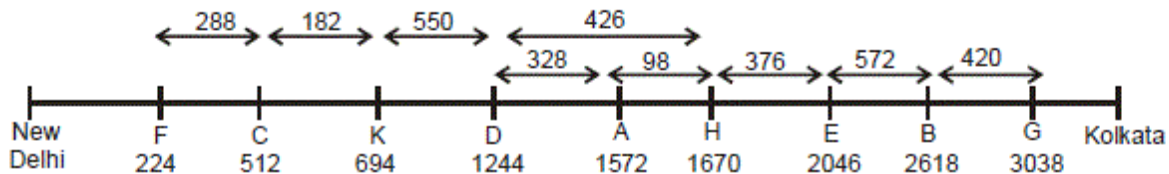


From 3rd row, we see the distances of B from E and G are 572 and 420 which adds upto 992 and in the above figure, the distance between E and G is 992, so B is in between E and G.

Similarly, from 4th row, the distances of C from F and K are 288 and 182 which adds upto 470 which is the distance between F and K, so C is in between F and K.

And also from row 5th, D is at distance 426 and 550 from H and K respectively, and the distance between H and K is 976 so D is in between H and K and we can also conclude it will be in between K and A.

So, finally we get,



If Kolkata is 568 km away from stoppage B, then that means the distance between Kolkata and stoppage G is 148 km.

Therefore, the distance between New Delhi and Kolkata is = $3038 + 148 = 3186$ km.

[Bookmark](#)

[FeedBack](#)

Directions for questions 31 to 34: Answer the questions on the basis of the information given below.

Five friends - Raj, Preeti, Romi, Sahib, and Aastha - each bought 1 T-shirt of different brands among six available brands of T-shirts in the shop – Roadster, Nike, Puma, Adidas, Incense, and Forever. Further information known to us:

- Aastha bought a T-shirt of neither Forever nor Roadster.
- Preeti bought a T-shirt of neither Adidas, nor Puma.
- If Raj bought T-shirt of brand Nike, then Romi bought one of Incense.
- Romi's T-shirt was costlier than Sahib's T-shirt, which was cheaper than Raj's T-shirt.
- Either Sahib's T-shirt was of brand Forever or Preeti's T-shirt was of brand Incense.
- Aastha's T-shirt was costlier than Preeti's T-shirt, which was costlier than only Roadster's T-shirt, which was bought by one of the 5 friends, but cheaper than Adidas T-shirt.
- The cheapest T-shirt was not of brand Forever, and the costliest T-shirt was of brand Nike. Only if Romi's T-shirt was of brand Puma, then Raj's T-shirt was not of brand Roadster.
- The Adidas T-shirt was costlier than Puma and Forever T-shirt, neither of which were bought by Raj.

Q.31 [11594329]

Who among the following bought the cheapest T-shirt?

1 ☐ Raj

2 ☐ Romi

3 ☐ Aastha

4 ☐ Sahib



Solution:

Correct Answer : 4

Your Answer : 4

Answer key/Solution

The given information can be summarised as follows:

From the information given, we can conclude the following information:

- (i) Aastha bought neither Roadster's nor Forever's T-shirt.
 - (ii) Preeti bought neither Adidas nor Puma's T-shirt.
 - (iii) If Raj bought Nike's, then Roman bought Incense's T-shirt.
 - (iv) Cost of Romi's T-shirt > cost of Sahib's T-shirt.
Cost of Raj's T-shirt > cost of Sahib's T-shirt.
 - (v) Either Sahib's T-shirt was Forever's or Preeti's T-shirt was Incense's.
 - (vi) Cost of Aastha's T-shirt > cost of Preeti's T-shirt, cost of Preeti's T-shirt > cost of Roadster T-shirt only.
And cost of Adidas T-shirt > cost of Preeti's T-shirt. Also, Roadster's t-shirt was bought by one of them.
 - (vii) Forever's t-shirt was not cheapest, and Nike's t-shirt was costliest.
 - (viii) Only if Romi's T-shirt was Puma then Raj's was not Roadster's.
 - (ix) Cost of Adidas T-shirt > cost of Puma and Forever.
 - (x) Raj bought neither Puma nor Forever.
- Using (iv) and (vi), we can conclude that Sahib T-shirt is Roadster, which is the cheapest of all other T-shirt and cost of Preeti's T-shirt was 4th highest.
 - Using (v), we can say that Preeti's T-shirt was of Incense.
 - Using (iii), we can say Raj did not buy Nike's.
 - Using (viii), it can be concluded that Romi's T-shirt was Puma's.
 - Using (vii) & (ix), we get,
Nike > Adidas > Puma / Forever.
 - Using (iii) & (ix), we can say Raj did not buy Nike's.
Hence, Aastha bought Nike's & Raj bought Adidas's.
Hence, we can conclude

Name	T-shirt Brand
Aastha	Niky
Raj	Adidas
Romi	Puma
Preeti	Incense
Sahib	Roadster

↑
Cost Increasing

Sahib bought the cheapest T-shirt.

Bookmark

Feedback

Directions for questions 31 to 34: Answer the questions on the basis of the information given below.

Five friends - Raj, Preeti, Romi, Sahib, and Aastha - each bought 1 T-shirt of different brands among six available brands of T-shirts in the shop – Roadster, Nike, Puma, Adidas, Incense, and Forever. Further information known to us:

- Aastha bought a T-shirt of neither Forever nor Roadster.
- Preeti bought a T-shirt of neither Adidas, nor Puma.
- If Raj bought T-shirt of brand Nike, then Romi bought one of Incense.
- Romi's T-shirt was costlier than Sahib's T-shirt, which was cheaper than Raj's T-shirt.
- Either Sahib's T-shirt was of brand Forever or Preeti's T-shirt was of brand Incense.
- Aastha's T-shirt was costlier than Preeti's T-shirt, which was costlier than only Roadster's T-shirt, which was bought by one of the 5 friends, but cheaper than Adidas T-shirt.
- The cheapest T-shirt was not of brand Forever, and the costliest T-shirt was of brand Nike. Only if Romi's T-shirt was of brand Puma, then Raj's T-shirt was not of brand Roadster.
- The Adidas T-shirt was costlier than Puma and Forever T-shirt, neither of which were bought by Raj.

Q.32 [11594329]

Who bought the costliest T-shirt?

1 ☐ Raj

2 ☐ Romi

3 ☐ Aastha

4 ☐ Sahib



Solution:

Correct Answer : 3

Your Answer : 3

Answer key/Solution

The given information can be summarised as follows:

From the information given, we can conclude the following information:

- (i) Aastha bought neither Roadster's nor Forever's T-shirt.
 - (ii) Preeti bought neither Adidas nor Puma's T-shirt.
 - (iii) If Raj bought Nike's, then Roman bought Incense's T-shirt.
 - (iv) Cost of Romi's T-shirt > cost of Sahib's T-shirt.
Cost of Raj's T-shirt > cost of Sahib's T-shirt.
 - (v) Either Sahib's T-shirt was Forever's or Preeti's T-shirt was Incense's.
 - (vi) Cost of Aastha's T-shirt > cost of Preeti's T-shirt, cost of Preeti's T-shirt > cost of Roadster T-shirt only.
And cost of Adidas T-shirt > cost of Preeti's T-shirt. Also, Roadster's t-shirt was bought by one of them.
 - (vii) Forever's t-shirt was not cheapest, and Nike's t-shirt was costliest.
 - (viii) Only if Romi's T-shirt was Puma then Raj's was not Roadster's.
 - (ix) Cost of Adidas T-shirt > cost of Puma and Forever.
 - (x) Raj bought neither Puma nor Forever.
- Using (iv) and (vi), we can conclude that Sahib T-shirt is Roadster, which is the cheapest of all other T-shirt and cost of Preeti's T-shirt was 4th highest.
 - Using (v), we can say that Preeti's T-shirt was of Incense.
 - Using (iii), we can say Raj did not buy Nike's.
 - Using (viii), it can be concluded that Romi's T-shirt was Puma's.
 - Using (vii) & (ix), we get,
Nike > Adidas > Puma / Forever.
 - Using (iii) & (ix), we can say Raj did not buy Nike's.
Hence, Aastha bought Nike's & Raj bought Adidas's.
Hence, we can conclude

Name	T-shirt Brand
Aastha	Niky
Raj	Adidas
Romi	Puma
Preeti	Incense
Sahib	Roadster

↑
Cost Increasing

Aastha bought the costliest T-shirt.

Bookmark

Feedback

Directions for questions 31 to 34: Answer the questions on the basis of the information given below.

Five friends - Raj, Preeti, Romi, Sahib, and Aastha - each bought 1 T-shirt of different brands among six available brands of T-shirts in the shop – Roadster, Nike, Puma, Adidas, Incense, and Forever. Further information known to us:

- Aastha bought a T-shirt of neither Forever nor Roadster.
- Preeti bought a T-shirt of neither Adidas, nor Puma.
- If Raj bought T-shirt of brand Nike, then Romi bought one of Incense.
- Romi's T-shirt was costlier than Sahib's T-shirt, which was cheaper than Raj's T-shirt.
- Either Sahib's T-shirt was of brand Forever or Preeti's T-shirt was of brand Incense.
- Aastha's T-shirt was costlier than Preeti's T-shirt, which was costlier than only Roadster's T-shirt, which was bought by one of the 5 friends, but cheaper than Adidas T-shirt.
- The cheapest T-shirt was not of brand Forever, and the costliest T-shirt was of brand Nike. Only if Romi's T-shirt was of brand Puma, then Raj's T-shirt was not of brand Roadster.
- The Adidas T-shirt was costlier than Puma and Forever T-shirt, neither of which were bought by Raj.

Q.33 [11594329]

The T-shirt of which brand was not bought by any of the 5 friends?

1 ☐ Roadster

2 ☐ Forever

3 ☐ Adidas

4 ☐ Puma



Solution:

Correct Answer : 2

Your Answer : 2

 **Answer key/Solution**

The given information can be summarised as follows:

From the information given, we can conclude the following information:

- (i) Aastha bought neither Roadster's nor Forever's T-shirt.
 - (ii) Preeti bought neither Adidas nor Puma's T-shirt.
 - (iii) If Raj bought Nike's, then Roman bought Incense's T-shirt.
 - (iv) Cost of Romi's T-shirt > cost of Sahib's T-shirt.
Cost of Raj's T-shirt > cost of Sahib's T-shirt.
 - (v) Either Sahib's T-shirt was Forever's or Preeti's T-shirt was Incense's.
 - (vi) Cost of Aastha's T-shirt > cost of Preeti's T-shirt, cost of Preeti's T-shirt > cost of Roadster T-shirt only.
And cost of Adidas T-shirt > cost of Preeti's T-shirt. Also, Roadster's t-shirt was bought by one of them.
 - (vii) Forever's t-shirt was not cheapest, and Nike's t-shirt was costliest.
 - (viii) Only if Romi's T-shirt was Puma then Raj's was not Roadster's.
 - (ix) Cost of Adidas T-shirt > cost of Puma and Forever.
 - (x) Raj bought neither Puma nor Forever.
- Using (iv) and (vi), we can conclude that Sahib T-shirt is Roadster, which is the cheapest of all other T-shirt and cost of Preeti's T-shirt was 4th highest.
 - Using (v), we can say that Preeti's T-shirt was of Incense.
 - Using (iii), we can say Raj did not buy Nike's.
 - Using (viii), it can be concluded that Romi's T-shirt was Puma's.
 - Using (vii) & (ix), we get,
Nike > Adidas > Puma / Forever.
 - Using (iii) & (ix), we can say Raj did not buy Nike's.
Hence, Aastha bought Nike's & Raj bought Adidas's.
Hence, we can conclude

Name	T-shirt Brand
Aastha	Niky
Raj	Adidas
Romi	Puma
Preeti	Incense
Sahib	Roadster

↑
Cost Increasing

Forever is the T-shirt of which was not bought by any of the 5 friends.

Bookmark

Feedback

Directions for questions 31 to 34: Answer the questions on the basis of the information given below.

Five friends - Raj, Preeti, Romi, Sahib, and Aastha - each bought 1 T-shirt of different brands among six available brands of T-shirts in the shop – Roadster, Nike, Puma, Adidas, Incense, and Forever. Further information known to us:

- Aastha bought a T-shirt of neither Forever nor Roadster.
- Preeti bought a T-shirt of neither Adidas, nor Puma.
- If Raj bought T-shirt of brand Nike, then Romi bought one of Incense.
- Romi's T-shirt was costlier than Sahib's T-shirt, which was cheaper than Raj's T-shirt.
- Either Sahib's T-shirt was of brand Forever or Preeti's T-shirt was of brand Incense.
- Aastha's T-shirt was costlier than Preeti's T-shirt, which was costlier than only Roadster's T-shirt, which was bought by one of the 5 friends, but cheaper than Adidas T-shirt.
- The cheapest T-shirt was not of brand Forever, and the costliest T-shirt was of brand Nike. Only if Romi's T-shirt was of brand Puma, then Raj's T-shirt was not of brand Roadster.
- The Adidas T-shirt was costlier than Puma and Forever T-shirt, neither of which were bought by Raj.

Q.34 [11594329]

Which of the following statements must be true?

- 1 ☐ Raj bought a Roadster's T-shirt.
- 2 ☐ Romi's T-shirt was costlier than Aastha's T-shirt.
- 3 ☐ Sahib bought Adidas T-shirt.
- 4 ☐ The Puma T-shirt was the 3rd costliest among the five T-shirts bought by the five friends.



Solution:

Correct Answer : 4

Your Answer : 4

Answer key/Solution

The given information can be summarised as follows:

From the information given, we can conclude the following information:

- (i) Aastha bought neither Roadster's nor Forever's T-shirt.
 - (ii) Preeti bought neither Adidas nor Puma's T-shirt.
 - (iii) If Raj bought Nike's, then Roman bought Incense's T-shirt.
 - (iv) Cost of Romi's T-shirt > cost of Sahib's T-shirt.
Cost of Raj's T-shirt > cost of Sahib's T-shirt.
 - (v) Either Sahib's T-shirt was Forever's or Preeti's T-shirt was Incense's.
 - (vi) Cost of Aastha's T-shirt > cost of Preeti's T-shirt, cost of Preeti's T-shirt > cost of Roadster T-shirt only.
And cost of Adidas T-shirt > cost of Preeti's T-shirt. Also, Roadster's t-shirt was bought by one of them.
 - (vii) Forever's t-shirt was not cheapest, and Nike's t-shirt was costliest.
 - (viii) Only if Romi's T-shirt was Puma then Raj's was not Roadster's.
 - (ix) Cost of Adidas T-shirt > cost of Puma and Forever.
 - (x) Raj bought neither Puma nor Forever.
- Using (iv) and (vi), we can conclude that Sahib T-shirt is Roadster, which is the cheapest of all other T-shirt and cost of Preeti's T-shirt was 4th highest.
 - Using (v), we can say that Preeti's T-shirt was of Incense.
 - Using (iii), we can say Raj did not buy Nike's.
 - Using (viii), it can be concluded that Romi's T-shirt was Puma's.
 - Using (vii) & (ix), we get,
Nike > Adidas > Puma / Forever.
 - Using (iii) & (ix), we can say Raj did not buy Nike's.
Hence, Aastha bought Nike's & Raj bought Adidas's.
Hence, we can conclude

Name	T-shirt Brand
Aastha	Niky
Raj	Adidas
Romi	Puma
Preeti	Incense
Sahib	Roadster

↑
Cost Increasing

Puma's is 3rd costliest t-shirt is true.

Bookmark

Feedback

Directions for questions 35 to 39: Answer the questions on the basis of the information given below.

A survey of 2000 people of a society was conducted by a company named XYZ Ltd. to gauge the popularity of three newspapers -TOI, The Hindu and The Indian Express. All of them voted for at least one newspaper. It is also known that:

- (i) The number of people who like TOI and The Hindu is 25% less than the people who like The Indian Express and The Hindu.
- (ii) The number of people who like only TOI is 50 more than 3 times the number of persons who like all newspapers.
- (iii) The number of people who like only The Indian Express is 50 less than 3 times the number of people who like only The Hindu.
- (iv) The number of people who like TOI and The Indian Express is double the number of people who like TOI and The Hindu.
- (v) The ratio of the number of people who like only The Indian Express to those who like TOI and The Indian Express but not The Hindu is 4 : 5.
- (vi) The number of people who like all three newspapers is one-third of the number of persons who like TOI and The Hindu.
- (vii) The number of persons who like only The Hindu is 70 more than one-fifth of one-fifth of the total number of people who participated.

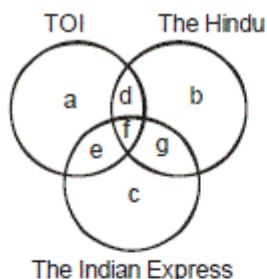
Q.35 [11594329]

What is the number of people who do not like TOI?

Solution:

Correct Answer : 850

 **Answer key/Solution**



As we know $a + b + c + d + e + f + g = 2000$

Statement (i): $d + f = 0.75(f + g) \Rightarrow d + f = \frac{3(f + g)}{4}$

Statement (ii): $a = 50 + 3f$

Statement (iii): $c = 3b - 50$

Statement (iv): $e + f = 2(d + f)$

Statement (v): $c : e = 4 : 5$

Statement (vi): $f = \frac{(d + f)}{3} \rightarrow 3f = d + f \Rightarrow d = 2f$

Statement (vii): $b = \frac{1}{5} \times \frac{1}{5} \times 2000 + 70 = 150$

From statement (ii): $c = 400$

From statement (v): $e = 500$

From statement (iv) and (vi): $500 + f = 2(2f + f) \Rightarrow 500 + f = 6f \Rightarrow f = 100$

From statement (vi): $d = 200$

From Statement (ii): $a = 350$

From statement (i): $200 + 100 = \frac{3(100 + g)}{4} \Rightarrow g = 300$

The people who like TOI are $a + d + f + e = 1150$.

So, the number of people who don't like TOI are $2000 - 1150 = 850$.

[Bookmark](#)
[FeedBack](#)

Directions for questions 35 to 39: Answer the questions on the basis of the information given below.

A survey of 2000 people of a society was conducted by a company named XYZ Ltd. to gauge the popularity of three newspapers -TOI, The Hindu and The Indian Express. All of them voted for at least one newspaper. It is also known that:

- (i) The number of people who like TOI and The Hindu is 25% less than the people who like The Indian Express and The Hindu.
- (ii) The number of people who like only TOI is 50 more than 3 times the number of persons who like all newspapers.
- (iii) The number of people who like only The Indian Express is 50 less than 3 times the number of people who like only The Hindu.
- (iv) The number of people who like TOI and The Indian Express is double the number of people who like TOI and The Hindu.
- (v) The ratio of the number of people who like only The Indian Express to those who like TOI and The Indian Express but not The Hindu is 4 : 5.
- (vi) The number of people who like all three newspapers is one-third of the number of persons who like TOI and The Hindu.
- (vii) The number of persons who like only The Hindu is 70 more than one-fifth of one-fifth of the total number of people who participated.

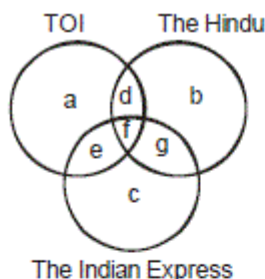
Q.36 [11594329]

What is the number of people who do not like TOI and The Hindu but like The Indian Express?

Solution:

Correct Answer : 400

 **Answer key/Solution**



As we know $a + b + c + d + e + f + g = 2000$

Statement (i): $d + f = 0.75(f + g) \Rightarrow d + f = \frac{3(f + g)}{4}$

Statement (ii): $a = 50 + 3f$

Statement (iii): $c = 3b - 50$

Statement (iv): $e + f = 2(d + f)$

Statement (v): $c : e = 4 : 5$

Statement (vi): $f = \frac{(d + f)}{3} \rightarrow 3f = d + f \Rightarrow d = 2f$

Statement (vii): $b = \frac{1}{5} \times \frac{1}{5} \times 2000 + 70 = 150$

From statement (ii): $c = 400$

From statement (v): $e = 500$

From statement (iv) and (vi): $500 + f = 2(2f + f) \Rightarrow 500 + f = 6f \Rightarrow f = 100$

From statement (vi): $d = 200$

From Statement (ii): $a = 350$

From statement (i): $200 + 100 = \frac{3(100 + g)}{4} \Rightarrow g = 300$

The number of people who do not like TOI and The Hindu but like The Indian Express will be only people who like only The Indian express i.e., 400.

[Bookmark](#)
[FeedBack](#)

Directions for questions 35 to 39: Answer the questions on the basis of the information given below.

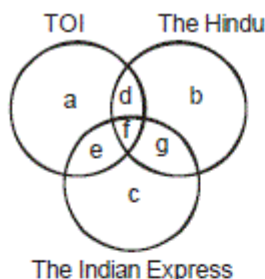
A survey of 2000 people of a society was conducted by a company named XYZ Ltd. to gauge the popularity of three newspapers -TOI, The Hindu and The Indian Express. All of them voted for at least one newspaper. It is also known that:

- (i) The number of people who like TOI and The Hindu is 25% less than the people who like The Indian Express and The Hindu.
- (ii) The number of people who like only TOI is 50 more than 3 times the number of persons who like all newspapers.
- (iii) The number of people who like only The Indian Express is 50 less than 3 times the number of people who like only The Hindu.
- (iv) The number of people who like TOI and The Indian Express is double the number of people who like TOI and The Hindu.
- (v) The ratio of the number of people who like only The Indian Express to those who like TOI and The Indian Express but not The Hindu is 4 : 5.
- (vi) The number of people who like all three newspapers is one-third of the number of persons who like TOI and The Hindu.
- (vii) The number of persons who like only The Hindu is 70 more than one-fifth of one-fifth of the total number of people who participated.

Q.37 [11594329]

If the number of people who likes only The Hindu and the people who likes only The Indian Express are interchanged, then which newspaper is liked by maximum number of people?

- 1 ☐ The Hindu
- 2 ☐ The Indian Express
- 3 ☐ TOI
- 4 ☐ Cannot be determined

Solution:**Correct Answer : 3**[Answer key/Solution](#)

As we know $a + b + c + d + e + f + g = 2000$

Statement (i): $d + f = 0.75(f + g) \Rightarrow d + f = \frac{3(f + g)}{4}$

Statement (ii): $a = 50 + 3f$

Statement (iii): $c = 3b - 50$

Statement (iv): $e + f = 2(d + f)$

Statement (v): $c : e = 4 : 5$

Statement (vi): $f = \frac{(d + f)}{3} \rightarrow 3f = d + f \Rightarrow d = 2f$

Statement (vii): $b = \frac{1}{5} \times \frac{1}{5} \times 2000 + 70 = 150$

From statement (ii): $c = 400$

From statement (v): $e = 500$

From statement (iv) and (vi): $500 + f = 2(2f + f) \Rightarrow 500 + f = 6f \Rightarrow f = 100$

From statement (vi): $d = 200$

From Statement (ii): $a = 350$

From statement (i): $200 + 100 = \frac{3(100 + g)}{4} \Rightarrow g = 300$

If b and c are interchanged, the people who like The Hindu will be =1000

The people who like The Indian Express will be =1050

The people who like TOI will be = 1150

So, correct answer will be TOI.

[Bookmark](#)
[FeedBack](#)

Directions for questions 35 to 39: Answer the questions on the basis of the information given below.

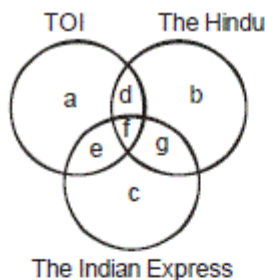
A survey of 2000 people of a society was conducted by a company named XYZ Ltd. to gauge the popularity of three newspapers -TOI, The Hindu and The Indian Express. All of them voted for at least one newspaper. It is also known that:

- (i) The number of people who like TOI and The Hindu is 25% less than the people who like The Indian Express and The Hindu.
- (ii) The number of people who like only TOI is 50 more than 3 times the number of persons who like all newspapers.
- (iii) The number of people who like only The Indian Express is 50 less than 3 times the number of people who like only The Hindu.
- (iv) The number of people who like TOI and The Indian Express is double the number of people who like TOI and The Hindu.
- (v) The ratio of the number of people who like only The Indian Express to those who like TOI and The Indian Express but not The Hindu is 4 : 5.
- (vi) The number of people who like all three newspapers is one-third of the number of persons who like TOI and The Hindu.
- (vii) The number of persons who like only The Hindu is 70 more than one-fifth of one-fifth of the total number of people who participated.

Q.38 [11594329]

If number of people who likes only The Hindu and the people who likes only The Indian Express are interchanged, then which newspapers likeliness increased by maximum percentage?

- 1 ☐ The Hindu
- 2 ☐ The Indian Express
- 3 ☐ TOI
- 4 ☐ Cannot be determined

Solution:**Correct Answer : 1**[Answer key/Solution](#)

As we know $a + b + c + d + e + f + g = 2000$

Statement (i): $d + f = 0.75(f + g) \Rightarrow d + f = \frac{3(f + g)}{4}$

Statement (ii): $a = 50 + 3f$

Statement (iii): $c = 3b - 50$

Statement (iv): $e + f = 2(d + f)$

Statement (v): $c : e = 4 : 5$

Statement (vi): $f = \frac{(d + f)}{3} \rightarrow 3f = d + f \Rightarrow d = 2f$

Statement (vii): $b = \frac{1}{5} \times \frac{1}{5} \times 2000 + 70 = 150$

From statement (ii): $c = 400$

From statement (v): $e = 500$

From statement (iv) and (vi): $500 + f = 2(2f + f) \Rightarrow 500 + f = 6f \Rightarrow f = 100$

From statement (vi): $d = 200$

From Statement (ii): $a = 350$

From statement (i): $200 + 100 = \frac{3(100 + g)}{4} \Rightarrow g = 300$

Newspaper	Initial Likelihood	Final Likelihood	Percentage Change
TOI	1150	1150	0
The Hindu	750	1050	40
The Indian Express	1300	1150	11.5

So, maximum percentage increase was in 'The Hindu'.

[Bookmark](#)
[FeedBack](#)

Directions for questions 35 to 39: Answer the questions on the basis of the information given below.

There are 13 students who are members of the cultural society of a B-School. Out of 13 students, 8 are boys – Aman, Barun, Chandan, Deepak, Eja, Fardeen, Gautam, and Hardik and rest five are girls – Princy, Raveena, Swarna, Ushi, and Vidushi. They are to get sponsorship for the annual cultural fest of the B-School. For this, they form three teams – A, B, and C - from among them. In each team, there are at least 3 students and at least one girl. Further, every student is a part of at least one team.

Following is the additional information:

- (i) No two students out of Aman, Barun, and Chandan are part of the same team.
- (ii) If Aman is part of any team, then Ushi is also part of the same team and vice versa while Princy and Swarna are never part of the same team.
- (iii) Eja is always a part of either Team A or Team B.
- (iv) Atleast one and atmost two out of Vidushi, Raveena, and Hardik must be a part of Team C.
- (v) Further, Princy is always a part of Team B only, Barun is never a part of Team C and Swarna is never a part of Team A.
- (vi) Whenever Swarna is part of any team, then Deepak is also part of the same team and vice versa.
- (vii) Each team must have at least 2 boys but not more than 2 girls.

Q.39 [11594329]

Which of the following could be a part of only Team C and of no other team?

1 ☐ Deepak

2 ☐ Eja

3 ☐ Hardik

4 ☐ Vidushi

✕

Solution:

Correct Answer : 1

Your Answer : 3

Using points (ii) and (v), since Princy and Swarna cannot be in the same team and

Princy is always in Team B and Swarna is never in Team A, therefore, Swarna can only be a part of Team C.

And using point (vi), it can be concluded that Deepak must be a part of only Team C.

Bookmark

FeedBack

 Answer key/Solution

Directions for questions 40 to 42: Answer the questions on the basis of the information given below.

There are 13 students who are members of the cultural society of a B-School. Out of 13 students, 8 are boys – Aman, Barun, Chandan, Deepak, Eja, Fardeen, Gautam, and Hardik and rest five are girls – Princy, Raveena, Swarna, Ushi, and Vidushi. They are to get sponsorship for the annual cultural fest of the B-School. For this, they form three teams – A, B, and C - from among them. In each team, there are at least 3 students and at least one girl. Further, every student is a part of at least one team.

Following is the additional information:

- (i) No two students out of Aman, Barun, and Chandan are part of the same team.
- (ii) If Aman is part of any team, then Ushi is also part of the same team and vice versa while Princy and Swarna are never part of the same team.
- (iii) Eja is always a part of either Team A or Team B.
- (iv) Atleast one and atmost two out of Vidushi, Raveena, and Hardik must be a part of Team C.
- (v) Further, Princy is always a part of Team B only, Barun is never a part of Team C and Swarna is never a part of Team A.
- (vi) Whenever Swarna is part of any team, then Deepak is also part of the same team and vice versa.
- (vii) Each team must have at least 2 boys but not more than 2 girls.

Q.40 [11594329]

What could be the maximum strength (in terms of number of students) of Team A, if no student is common in any team?

1 ☐ 7

2 ☐ 6

3 ☐ 5

4 ☐ 4

✕

Solution:

Correct Answer : 2

Your Answer : 4

 Answer key/Solution

Since no student is common in any of the 2 teams and we have to maximize the strength of Team A, therefore, we need to minimize the other two teams by making them the teams of 3 students each.

One such possible scenario is as follows:

Team C $\Rightarrow$ Chandan, Swarna, Deepak, and Raveena; (Remember atleast one of Hardik, Vidushi and Raveena must be in Team C).

Team B $\Rightarrow$ Princy, Barun, Eja;

Team A $\Rightarrow$ Aman, Fardeen, Gautam, Hardik, Ushi, and Vidushi.

Note: In Team C, either one of Raveena or Vidhushi should be given, but not Hardik. As there is a limitation on the member of girls in any team but there is no upper limit on the number of boys in the team, therefore, Team A can accomodate any number of boys but not more than 2 girls.

Bookmark

FeedBack

Directions for questions 40 to 42: Answer the questions on the basis of the information given below.

There are 13 students who are members of the cultural society of a B-School. Out of 13 students, 8 are boys – Aman, Barun, Chandan, Deepak, Eja, Fardeen, Gautam, and Hardik and rest five are girls – Princy, Raveena, Swarna, Ushi, and Vidushi. They are to get sponsorship for the annual cultural fest of the B-School. For this, they form three teams – A, B, and C - from among them. In each team, there are at least 3 students and at least one girl. Further, every student is a part of at least one team.

Following is the additional information:

- (i) No two students out of Aman, Barun, and Chandan are part of the same team.
- (ii) If Aman is part of any team, then Ushi is also part of the same team and vice versa while Princy and Swarna are never part of the same team.
- (iii) Eja is always a part of either Team A or Team B.
- (iv) Atleast one and atmost two out of Vidushi, Raveena, and Hardik must be a part of Team C.
- (v) Further, Princy is always a part of Team B only, Barun is never a part of Team C and Swarna is never a part of Team A.
- (vi) Whenever Swarna is part of any team, then Deepak is also part of the same team and vice versa.
- (vii) Each team must have at least 2 boys but not more than 2 girls.

Q.41 [11594329]

If Aman, Ushi, and Raveena are part of Team A and Princy, Barun, and Fardeen are part of Team B, then how many different compositions of Team C are possible if Vidushi and Hardik are definitely part of Team C?

1 ☐ 15

2 ☐ 10

3 ☐ 9

4 ☐ 8

Solution:**Correct Answer : 4** Answer key/Solution

If Aman, Ushi, and Raveena are part of Team A and Princy, Barun, and Fardeen are part of Team B,

Team A	Team A	Team A
Aman	Princy	Chandan
Ushi	Barun	Swarna
Raveena	Fardeen	Deepak
		Vidushi
		Hardik

In order to find the possible compositions for team C, observe all those students who can be in team C other than the 5 mentioned students.

Fardeen, Gautam, and Eja (Eja will be considered because point (iii) says, he will always be either in team A or B, but it doesn't tell if he is a part of team C or not, therefore he could be a part of team C).

One possible combination is Chandan Swarna, Deepak, Vidushi, and Hardik. And other will be formed by taking all possible combinations of Fardeen, Gautam, and Eja, which are 7.

Total possible combinations are = $1 + 7 = 8$.

Bookmark

FeedBack

Directions for questions 40 to 42: Answer the questions on the basis of the information given below.

There are 13 students who are members of the cultural society of a B-School. Out of 13 students, 8 are boys – Aman, Barun, Chandan, Deepak, Eja, Fardeen, Gautam, and Hardik and rest five are girls – Princy, Raveena, Swarna, Ushi, and Vidushi. They are to get sponsorship for the annual cultural fest of the B-School. For this, they form three teams – A, B, and C - from among them. In each team, there are at least 3 students and at least one girl. Further, every student is a part of at least one team.

Following is the additional information:

- (i) No two students out of Aman, Barun, and Chandan are part of the same team.
- (ii) If Aman is part of any team, then Ushi is also part of the same team and vice versa while Princy and Swarna are never part of the same team.
- (iii) Eja is always a part of either Team A or Team B.
- (iv) Atleast one and atmost two out of Vidushi, Raveena, and Hardik must be a part of Team C.
- (v) Further, Princy is always a part of Team B only, Barun is never a part of Team C and Swarna is never a part of Team A.
- (vi) Whenever Swarna is part of any team, then Deepak is also part of the same team and vice versa.
- (vii) Each team must have at least 2 boys but not more than 2 girls.

Q.42 [11594329]

If Team A comprise of only 4 students – Aman, Ushi, Fardeen, and Gautam, then how many different composition of Team B are possible?

1 ☐ 92 ☐ 12

3 ○ 16

4 ○ 8

Solution:**Correct Answer : 3** Answer key/Solution

If Aman is in Team A, then Barun must be a part of Team B. (Refer pts. (i) and (iii)). Since Eja is not in Team A therefore, he must be in Team B (Refer pt. (iii)) and Princy is always in Team B (Refer pt. (v)). Now out of the remaining students, following are the possibilities:

Deepak → Must be in Team C only (Refer pts. (ii) and (vi))

Hardik → 2 options → may or may not be in Team B

Fardeen → Same as Hardik

Gautam → Same as Hardik

Swarna → Always in Team C

Ushi → Always in team C (Refer pt. (ii))

Now, out of Vidushi and Raveena, at most one can be a part of Team B (Refer pt. (iv)) and hence 2 options (i.e., only Vidushi, or only Raveena).

Hence, total compositions for Team B

$$= \underset{\text{Hardik}}{2} \times \underset{\text{Gautam}}{2} \times \underset{\text{Fardeen}}{2} \times \underset{\text{Vidushi / Raveena}}{2} = 16.$$

Bookmark

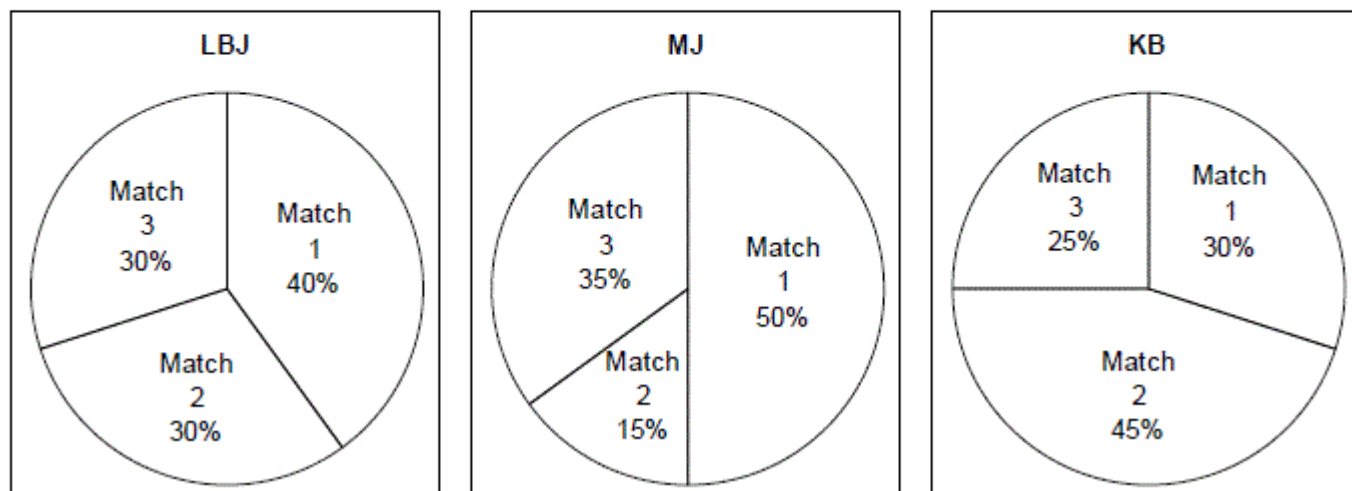
FeedBack

Directions for questions 43 to 46: Answer the questions on the basis of the information given below.

NBL championship is a very famous basketball league which has three stages - I, II and III. Three top players LeBron James (LBJ), Michael Jordon (MJ) and Kobe Bryant (KB) played in this championship. LBJ played only in stage I, MJ in stage II and KB in stage III. In a match, if a player puts the basketball in the opposition's basket we call it a "pointer" and depending on the distance from where the basketball is put in the basket, points are awarded. There are three different types of pointers - 2 pointers, 3 pointers and 4 pointers. The coach of the players mentioned above, maintains a log of how many pointers and of which type is scored by the three players and awards them 3 marks for a 3 pointer and 4 marks for a 4 pointer. (No marks are awarded for 2 pointers). The following table provides us the information about the percentage wise distribution of 3 pointers and 4 pointers in the three matches of every stage.

	Match 1	Match 2	Match-3
3-pointers	40%	35%	25%
4-pointers	20%	30%	50%

The following pie charts give the match-wise distribution of marks awarded to the players by their coach:



The total number of 3 pointers and 4 pointers scored in these three matches of every stage is 400, out of whom 100 pointers were scored in Match 1 of every stage. Total marks awarded by the coach for all three matches combined to LBJ are 500, to MJ are 600 and to KB are 400.

Q.43 [11594329]

What could be the minimum number of 3 pointers scored by MJ in these three matches?

Solution:**Correct Answer : 24** Answer key/Solution

We know the total marks of LBJ, MJ, and KB and also their percentage wise break up scores in their three matches, so we can form the following table:

	Match 1	Match 2	Match 3	Total
LBJ	200	150	150	500
MJ	300	90	210	600
KB	120	180	100	400

Let the number of 3 pointers = x and the number of 4 pointers = y

Thus, $x + y = 400$ and 40% of $x + 20\%$ of $y = 100$ i.e., $\frac{2}{5}x + \frac{1}{5}y = 100$

Solving these two we get, $x = 100$ and $y = 300$.

Based on this we can form table for number of 3 pointers and 4 pointers in three matches of every stage:

	Match 1	Match 2	Match 3	Total
3 pointers	40	35	25	100
4 pointers	60	90	150	300

In match 1, MJ can score all 60, 4-pointers and yet out of his total of 300, 60 marks still has to be scored through 3-pointers. Thus, he has to score 20, 3-pointers in match 1. In match 2 his score is 90 which is not a multiple of 4 so to minimise number of 3-pointers, we have to maximise the number of 4 pointers which will be 21 in this case and thus, the number of 3-pointers comes out to be 2. Similarly, in match 3, his score is 210 which is not a multiple of 4 so, to minimise number of 3-pointers we have to maximise the number of 4 pointers which will be 51 in this case and thus, the number of 3-pointers comes out to be 2. Thus, the minimum number of 3-pointers scored by him in three matches will be $20 + 2 + 2 = 24$.

Bookmark

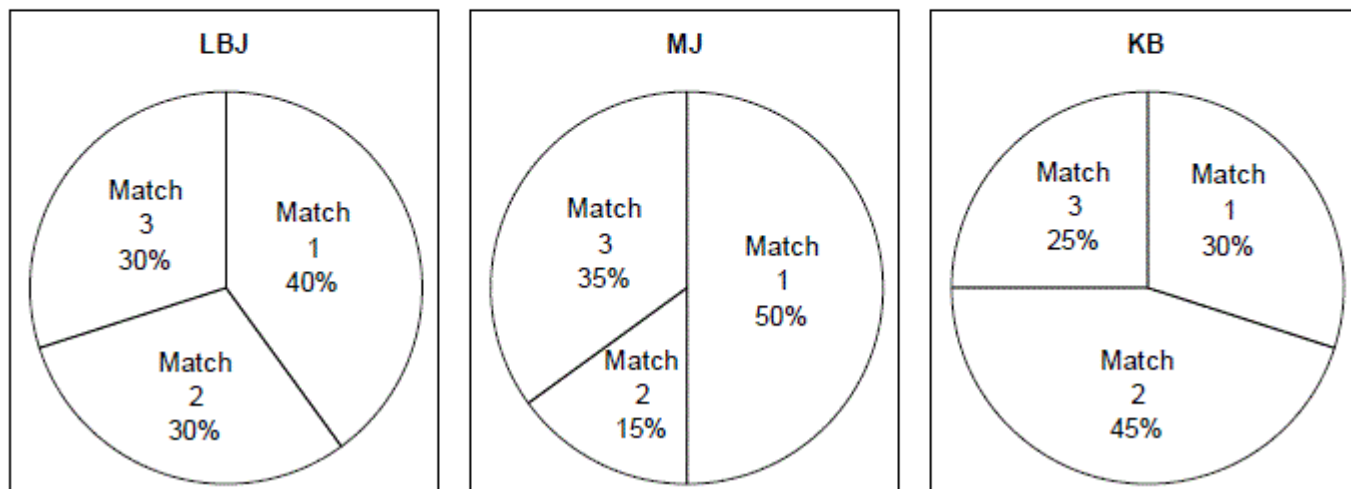
FeedBack

Directions for questions 43 to 46: Answer the questions on the basis of the information given below.

NBL championship is a very famous basketball league which has three stages - I, II and III. Three top players LeBron James (LBJ), Michael Jordan (MJ) and Kobe Bryant (KB) played in this championship. LBJ played only in stage I, MJ in stage II and KB in stage III. In a match, if a player puts the basketball in the opposition's basket we call it a "pointer" and depending on the distance from where the basketball is put in the basket, points are awarded. There are three different types of pointers - 2 pointers, 3 pointers and 4 pointers. The coach of the players mentioned above, maintains a log of how many pointers and of which type is scored by the three players and awards them 3 marks for a 3 pointer and 4 marks for a 4 pointer. (No marks are awarded for 2 pointers). The following table provides us the information about the percentage wise distribution of 3 pointers and 4 pointers in the three matches of every stage.

	Match 1	Match 2	Match-3
3-pointers	40%	35%	25%
4-pointers	20%	30%	50%

The following pie charts give the match-wise distribution of marks awarded to the players by their coach:



The total number of 3 pointers and 4 pointers scored in these three matches of every stage is 400, out of whom 100 pointers were scored in Match 1 of every stage. Total marks awarded by the coach for all three matches combined to LBJ are 500, to MJ are 600 and to KB are 400.

Q.44 [11594329]

If KB has scored minimum number of 4 pointers in match 2, then what is the ratio of number of 3 pointers and 4 pointers scored by him in match 2?

1 ☐ 32 : 21

2 ☐ 22 : 19

3 ☐ 39 : 23

4 ☐ 24 : 17

Solution:**Correct Answer : 1** Answer key/Solution

We know the total marks of LBJ, MJ, and KB and also their percentage wise break up scores in their three matches, so we can form the following table:

	Match 1	Match 2	Match 3	Total
LBJ	200	150	150	500
MJ	300	90	210	600
KB	120	180	100	400

Let the number of 3 pointers = x and the number of 4 pointers = y

Thus, $x + y = 400$ and 40% of $x + 20\%$ of $y = 100$ i.e., $\frac{2}{5}x + \frac{1}{5}y = 100$

Solving these two we get, $x = 100$ and $y = 300$.

Based on this we can form table for number of 3 pointers and 4 pointers in three matches of every stage:

	Match 1	Match 2	Match 3	Total
3 pointers	40	35	25	100
4 pointers	60	90	150	300

As KB has scored minimum number of 4-pointers in match 2, we have to maximise the number of 3-pointers. The maximum number of 3-pointers he can score is 35 but then the remaining marks (i.e., $180 - (35 \times 3) = 75$) will not be scored through 4-pointers as it is not divisible by 4. So, the possible value of number of 3-pointers in this case will be 32 and thus, the number of 4-pointers comes out to be 21. Hence. the required ratio is 32 : 21.

Bookmark

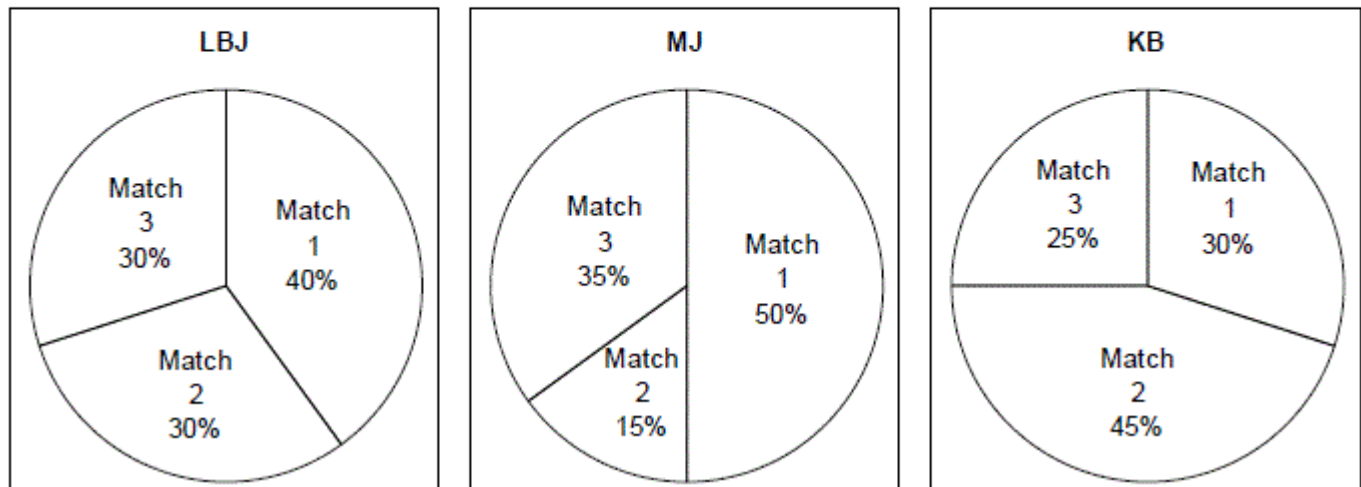
FeedBack

Directions for questions 43 to 46: Answer the questions on the basis of the information given below.

NBL championship is a very famous basketball league which has three stages - I, II and III. Three top players LeBron James (LBJ), Michael Jordan (MJ) and Kobe Bryant (KB) played in this championship. LBJ played only in stage I, MJ in stage II and KB in stage III. In a match, if a player puts the basketball in the opposition's basket we call it a "pointer" and depending on the distance from where the basketball is put in the basket, points are awarded. There are three different types of pointers - 2 pointers, 3 pointers and 4 pointers. The coach of the players mentioned above, maintains a log of how many pointers and of which type is scored by the three players and awards them 3 marks for a 3 pointer and 4 marks for a 4 pointer. (No marks are awarded for 2 pointers). The following table provides us the information about the percentage wise distribution of 3 pointers and 4 pointers in the three matches of every stage.

	Match 1	Match 2	Match-3
3-pointers	40%	35%	25%
4-pointers	20%	30%	50%

The following pie charts give the match-wise distribution of marks awarded to the players by their coach:



The total number of 3 pointers and 4 pointers scored in these three matches of every stage is 400, out of whom 100 pointers were scored in Match 1 of every stage. Total marks awarded by the coach for all three matches combined to LBJ are 500, to MJ are 600 and to KB are 400.

Q.45 [11594329]

What can be the number of possible values of the number of 3 pointers that LBJ could have scored in match 2?

1 ☐ 13

2 ☐ 9

3 ☐ 12

4 ☐ 10

Solution:**Correct Answer : 2** Answer key/Solution

We know the total marks of LBJ, MJ, and KB and also their percentage wise break up scores in their three matches, so we can form the following table:

	Match 1	Match 2	Match 3	Total
LBJ	200	150	150	500
MJ	300	90	210	600
KB	120	180	100	400

Let the number of 3 pointers = x and the number of 4 pointers = y

Thus, $x + y = 400$ and 40% of $x + 20\%$ of $y = 100$ i.e., $\frac{2}{5}x + \frac{1}{5}y = 100$

Solving these two we get, $x = 100$ and $y = 300$.

Based on this we can form table for number of 3 pointers and 4 pointers in three matches of every stage:

	Match 1	Match 2	Match 3	Total
3 pointers	40	35	25	100
4 pointers	60	90	150	300

Let us assume LBJ scored 'p' number of 3-pointers and 'q' number of 4-pointers in match 2. Thus, $3p + 4q = 150$. We have to find the whole number solution to this equation.

$q = 0, p = 50$; $q = 3, p = 46$; $q = 6, p = 42$; $q = 9, p = 38$; $q = 12, p = 34$; $q = 15, p = 30$; $q = 18, p = 26$; $q = 21, p = 22$; $q = 24, p = 18$; $q = 27, p = 14$; $q = 30, p = 10$; $q = 33, p = 6$; $q = 36, p = 2$. These are 13 values. But in match 2, the number of 3-pointers cannot be more than 35. So, the starting 4 values will not be included. Thus, the number of possible values = 9.

Bookmark

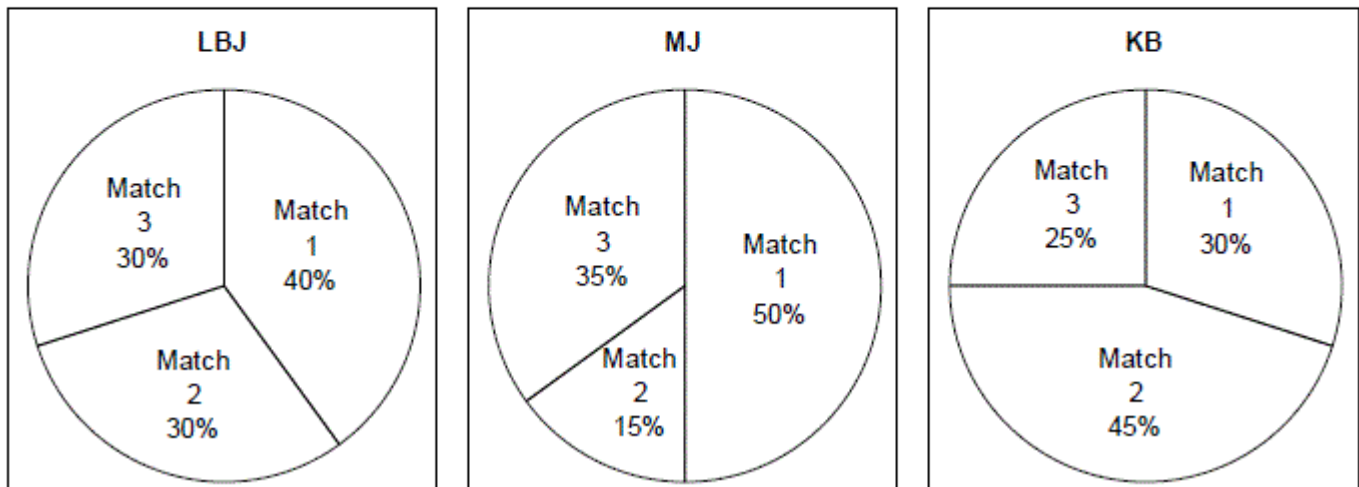
FeedBack

Directions for questions 43 to 46: Answer the questions on the basis of the information given below.

NBL championship is a very famous basketball league which has three stages - I, II and III. Three top players LeBron James (LBJ), Michael Jordon (MJ) and Kobe Bryant (KB) played in this championship. LBJ played only in stage I, MJ in stage II and KB in stage III. In a match, if a player puts the basketball in the opposition's basket we call it a "pointer" and depending on the distance from where the basketball is put in the basket, points are awarded. There are three different types of pointers - 2 pointers, 3 pointers and 4 pointers. The coach of the players mentioned above, maintains a log of how many pointers and of which type is scored by the three players and awards them 3 marks for a 3 pointer and 4 marks for a 4 pointer. (No marks are awarded for 2 pointers). The following table provides us the information about the percentage wise distribution of 3 pointers and 4 pointers in the three matches of every stage.

	Match 1	Match 2	Match-3
3-pointers	40%	35%	25%
4-pointers	20%	30%	50%

The following pie charts give the match-wise distribution of marks awarded to the players by their coach:



The total number of 3 pointers and 4 pointers scored in these three matches of every stage is 400, out of whom 100 pointers were scored in Match 1 of every stage. Total marks awarded by the coach for all three matches combined to LBJ are 500, to MJ are 600 and to KB are 400.

Q.46 [11594329]

Find the ratio of the minimum number of 3 pointers scored by LBJ in match 3 to the maximum number of 4 pointers scored by MJ in match 2.

1 ☐ 3 : 16

2 ☐ 2 : 19

3 ☐ 2 : 21

4 ☐ 4 : 15

Solution:**Correct Answer : 3** Answer key/Solution

We know the total marks of LBJ, MJ, and KB and also their percentage wise break up scores in their three matches, so we can form the following table:

	Match 1	Match 2	Match 3	Total
LBJ	200	150	150	500
MJ	300	90	210	600
KB	120	180	100	400

Let the number of 3 pointers = x and the number of 4 pointers = y

Thus, $x + y = 400$ and 40% of x + 20% of y = 100 i.e., $\frac{2}{5}x + \frac{1}{5}y = 100$

Solving these two we get, x = 100 and y = 300.

Based on this we can form table for number of 3 pointers and 4 pointers in three matches of every stage:

	Match 1	Match 2	Match 3	Total
3 pointers	40	35	25	100
4 pointers	60	90	150	300

Let LBJ scored 'a' number of 3-pointers and 'b' number of 4-pointers in match 3. Thus, $3a + 4b = 150$, so the minimum value of 'a' that satisfies is 2.

Let MJ scored 'p' number of 3-pointers and 'q' number of 4-pointers in match 2. Thus, $3p + 4q = 90$, so the maximum value of q that satisfies is 21.

Therefore, the required ratio = 2 : 21.

Bookmark

FeedBack

Directions for questions 47 to 50: Answer the questions on the basis of the information given below.

Three friends - Ted, Marshal, and Barney have a hobby of collecting Postal Stamps. It is known that the number of Postal Stamps with each of them is between 83 to 89 (both inclusive). It is known that exactly one of the friends always tells the truth, exactly one always tell a lie and the remaining one alternates between truth and lie in any order. It is also known that the number of Postal Stamps with the three friends is distinct and is an integral value. The following are the statements made by the three friends:

Ted

1. The number of Postal Stamps with me is 2 more than that with Barney.
2. Marshal is not telling the correct number of Postal Stamps with him.
3. The number of Postal Stamps with Barney is between 84 and 88 (both inclusive).

Marshal

1. The number of Postal Stamps with me is 84.
2. The number of Postal Stamps with Barney is 86.
3. The number of Postal Stamps with Ted is lesser than me.

Barney

1. Ted does not have the most number of Postal Stamps.
2. The number of Postal Stamps with me is 89.
3. Ted is lying about the number of Postal Stamps with me.

Q.47 [11594329]

If Marshal is the alternator, then which of the following is true?

- 1 ☐ The number of Postal Stamps with Barney is 89.
- 2 ☐ The number of Postal Stamps with Barney is 86.
- 3 ☐ Neither (1) nor (2)
- 4 ☐ Either (1) or (2)

Solution:

Correct Answer : 4

 Answer key/Solution

All the questions asked are condition based so let us make cases:

Case 1: Ted is the truth teller. Now all his statements are true. This will make Marshal's 1st statement false but the 2nd statement true. Thus, he becomes the alternator and his 3rd statement becomes false. Now Barney has to be liar. Now let us check for any discrepancy and also try to evaluate the number of Postal Stamps with each one of them. Now, with 2nd statement of Marshal, 1st statement of Ted and 1st statement of Barney we can say that the number of Postal Stamps with Ted is highest and is 88 and the number of Postal Stamps with Barney is 86. We are not clear on what the number of Postal Stamps Marshal has but it can't be 84 and 89. We can make a table for this case:

Nature	Name	Number of Postal Stamps	Statement 1	Statement 2	Statement 3
Truth Teller	Ted	88	T	T	T
Alternator	Marshal	(not 84 or 89)	F	T	F
Liar	Barney	86	F	F	F

Case 2: Marshal is the truth teller. Now, Ted can either be a liar or an alternator, but his first two statements are false and the last statement becomes true. So this case does not hold true.

Case 3: Barney is the truth teller. Now, Marshal's second statement becomes false. So we can make two further cases:

Case A: Marshal is an alternator and Ted is the liar. Marshal's 1st and 3rd statements will be true. Considering this, Ted's number of Postal Stamps will be 83, Marshal's will be 84 and Barney's will be 89.

Nature	Name	Number of Postal Stamps	Statement 1	Statement 2	Statement 3
Liar	Ted	83	F	F	F
Alternator	Marshal	84	T	F	T
Truth Teller	Barney	89	T	T	T

Case B: Marshal is a liar and Ted is an alternator. Now Barney's 3rd statement is true so Ted's 1st and 3rd statement becomes false. Regarding the number of Postal Stamps we can deduce that only for Barney i.e., 89, for Marshal we can say it can't be 84 and will be less than that of Ted's.

Nature	Name	Number of Postal Stamps	Statement 1	Statement 2	Statement 3
Alternator	Ted		F	T	F
Liar	Marshal	(<Ted's, not 84)	F	F	F
Truth Teller	Barney	89	T	T	T

If Marshal is an alternator, then the number of postal stamps with Barney is either 89 or 86.

[Bookmark](#)
[FeedBack](#)

Directions for questions 47 to 50: Answer the questions on the basis of the information given below.

Three friends - Ted, Marshal, and Barney have a hobby of collecting Postal Stamps. It is known that the number of Postal Stamps with each of them is between 83 to 89 (both inclusive). It is known that exactly one of the friends always tells the truth, exactly one always tell a lie and the remaining one alternates between truth and lie in any order. It is also known that the number of Postal Stamps with the three friends is distinct and is an integral value. The following are the statements made by the three friends:

Ted

1. The number of Postal Stamps with me is 2 more than that with Barney.
2. Marshal is not telling the correct number of Postal Stamps with him.
3. The number of Postal Stamps with Barney is between 84 and 88 (both inclusive).

Marshal

1. The number of Postal Stamps with me is 84.
2. The number of Postal Stamps with Barney is 86.
3. The number of Postal Stamps with Ted is lesser than me.

Barney

1. Ted does not have the most number of Postal Stamps.
2. The number of Postal Stamps with me is 89.
3. Ted is lying about the number of Postal Stamps with me.

Q.48 [11594329]

Which of the following is not possible?

- 1 ☐ Ted always lies.
- 2 ☐ Barney always lies.
- 3 ☐ Marshal always tells the truth.
- 4 ☐ Barney always tells the truth.

Solution:

Correct Answer : 3

 Answer key/Solution

All the questions asked are condition based so let us make cases:

Case 1: Ted is the truth teller. Now all his statements are true. This will make Marshal's 1st statement false but the 2nd statement true. Thus, he becomes the alternator and his 3rd statement becomes false. Now Barney has to be liar. Now let us check for any discrepancy and also try to evaluate the number of Postal Stamps with each one of them. Now, with 2nd statement of Marshal, 1st statement of Ted and 1st statement of Barney we can say that the number of Postal Stamps with Ted is highest and is 88 and the number of Postal Stamps with Barney is 86. We are not clear on what the number of Postal Stamps Marshal has but it can't be 84 and 89. We can make a table for this case:

Nature	Name	Number of Postal Stamps	Statement 1	Statement 2	Statement 3
Truth Teller	Ted	88	T	T	T
Alternator	Marshal	(not 84 or 89)	F	T	F
Liar	Barney	86	F	F	F

Case 2: Marshal is the truth teller. Now, Ted can either be a liar or an alternator, but his first two statements are false and the last statement becomes true. So this case does not hold true.

Case 3: Barney is the truth teller. Now, Marshal's second statement becomes false. So we can make two further cases:

Case A: Marshal is an alternator and Ted is the liar. Marshal's 1st and 3rd statements will be true. Considering this, Ted's number of Postal Stamps will be 83, Marshal's will be 84 and Barney's will be 89.

Nature	Name	Number of Postal Stamps	Statement 1	Statement 2	Statement 3
Liar	Ted	83	F	F	F
Alternator	Marshal	84	T	F	T
Truth Teller	Barney	89	T	T	T

Case B: Marshal is a liar and Ted is an alternator. Now Barney's 3rd statement is true so Ted's 1st and 3rd statement becomes false. Regarding the number of Postal Stamps we can deduce that only for Barney i.e., 89, for Marshal we can say it can't be 84 and will be less than that of Ted's.

Nature	Name	Number of Postal Stamps	Statement 1	Statement 2	Statement 3
Alternator	Ted		F	T	F
Liar	Marshal	(<Ted's, not 84)	F	F	F
Truth Teller	Barney	89	T	T	T

Among the given options, only third option is not possible.

[Bookmark](#)
[FeedBack](#)

Directions for questions 47 to 50: Answer the questions on the basis of the information given below.

Three friends - Ted, Marshal, and Barney have a hobby of collecting Postal Stamps. It is known that the number of Postal Stamps with each of them is between 83 to 89 (both inclusive). It is known that exactly one of the friends always tells the truth, exactly one always tell a lie and the remaining one alternates between truth and lie in any order. It is also known that the number of Postal Stamps with the three friends is distinct and is an integral value. The following are the statements made by the three friends:

Ted

1. The number of Postal Stamps with me is 2 more than that with Barney.
2. Marshal is not telling the correct number of Postal Stamps with him.
3. The number of Postal Stamps with Barney is between 84 and 88 (both inclusive).

Marshal

1. The number of Postal Stamps with me is 84.
2. The number of Postal Stamps with Barney is 86.
3. The number of Postal Stamps with Ted is lesser than me.

Barney

1. Ted does not have the most number of Postal Stamps.
2. The number of Postal Stamps with me is 89.
3. Ted is lying about the number of Postal Stamps with me.

Q.49 [11594329]

If Ted always speaks the truth, then how many of the following statements is/are definitely true?

- I. The number of Postal Stamps with Barney is 85.
- II. Ted does not have the highest number of Postal Stamps.
- III. The number of Postal Stamps with Marshal is 84.

1 ☐ 1

2 ☐ 0

3 ☐ 2

4 ☐ 3

Solution:**Correct Answer : 2**[Answer key/Solution](#)

All the questions asked are condition based so let us make cases:

Case 1: Ted is the truth teller. Now all his statements are true. This will make Marshal's 1st statement false but the 2nd statement true. Thus, he becomes the alternator and his 3rd statement becomes false. Now Barney has to be liar. Now let us check for any discrepancy and also try to evaluate the number of Postal Stamps with each one of them. Now, with 2nd statement of Marshal, 1st statement of Ted and 1st statement of Barney we can say that the number of Postal Stamps with Ted is highest and is 88 and the number of Postal Stamps with Barney is 86. We are not clear on what the number of Postal Stamps Marshal has but it can't be 84 and 89. We can make a table for this case:

Nature	Name	Number of Postal Stamps	Statement 1	Statement 2	Statement 3
Truth Teller	Ted	88	T	T	T
Alternator	Marshal	(not 84 or 89)	F	T	F
Liar	Barney	86	F	F	F

Case 2: Marshal is the truth teller. Now, Ted can either be a liar or an alternator, but his first two statements are false and the last statement becomes true. So this case does not hold true.

Case 3: Barney is the truth teller. Now, Marshal's second statement becomes false. So we can make two further cases:

Case A: Marshal is an alternator and Ted is the liar. Marshal's 1st and 3rd statements will be true. Considering this, Ted's number of Postal Stamps will be 83, Marshal's will be 84 and Barney's will be 89.

Nature	Name	Number of Postal Stamps	Statement 1	Statement 2	Statement 3
Liar	Ted	83	F	F	F
Alternator	Marshal	84	T	F	T
Truth Teller	Barney	89	T	T	T

Case B: Marshal is a liar and Ted is an alternator. Now Barney's 3rd statement is true so Ted's 1st and 3rd statement becomes false. Regarding the number of Postal Stamps we can deduce that only for Barney i.e., 89, for Marshal we can say it can't be 84 and will be less than that of Ted's.

Nature	Name	Number of Postal Stamps	Statement 1	Statement 2	Statement 3
Alternator	Ted		F	T	F
Liar	Marshal	(<Ted's, not 84)	F	F	F
Truth Teller	Barney	89	T	T	T

If Ted is the truth teller, then none of the given options is definitely true.

[Bookmark](#)[FeedBack](#)

Directions for questions 47 to 50: Answer the questions on the basis of the information given below.

Three friends - Ted, Marshal, and Barney have a hobby of collecting Postal Stamps. It is known that the number of Postal Stamps with each of them is between 83 to 89 (both inclusive). It is known that exactly one of the friends always tells the truth, exactly one always tell a lie and the remaining one alternates between truth and lie in any order. It is also known that the number of Postal Stamps with the three friends is distinct and is an integral value. The following are the statements made by the three friends:

Ted

1. The number of Postal Stamps with me is 2 more than that with Barney.
2. Marshal is not telling the correct number of Postal Stamps with him.
3. The number of Postal Stamps with Barney is between 84 and 88 (both inclusive).

Marshal

1. The number of Postal Stamps with me is 84.
2. The number of Postal Stamps with Barney is 86.
3. The number of Postal Stamps with Ted is lesser than me.

Barney

1. Ted does not have the most number of Postal Stamps.
2. The number of Postal Stamps with me is 89.
3. Ted is lying about the number of Postal Stamps with me.

Q.50 [11594329]

The number of Postal Stamps with Barney is 89 when

- I. Ted is a liar.
- II. Marshal is a liar.

1 ☐ Only I

2 ☐ Only II

3 ☐ Either I or II

4 ☐ Neither I nor II

Solution:**Correct Answer : 3**[Answer key/Solution](#)

All the questions asked are condition based so let us make cases:

Case 1: Ted is the truth teller. Now all his statements are true. This will make Marshal's 1st statement false but the 2nd statement true. Thus, he becomes the alternator and his 3rd statement becomes false. Now Barney has to be liar. Now let us check for any discrepancy and also try to evaluate the number of Postal Stamps with each one of them. Now, with 2nd statement of Marshal, 1st statement of Ted and 1st statement of Barney we can say that the number of Postal Stamps with Ted is highest and is 88 and the number of Postal Stamps with Barney is 86. We are not clear on what the number of Postal Stamps Marshal has but it can't be 84 and 89. We can make a table for this case:

Nature	Name	Number of Postal Stamps	Statement 1	Statement 2	Statement 3
Truth Teller	Ted	88	T	T	T
Alternator	Marshal	(not 84 or 89)	F	T	F
Liar	Barney	86	F	F	F

Case 2: Marshal is the truth teller. Now, Ted can either be a liar or an alternator, but his first two statements are false and the last statement becomes true. So this case does not hold true.

Case 3: Barney is the truth teller. Now, Marshal's second statement becomes false. So we can make two further cases:

Case A: Marshal is an alternator and Ted is the liar. Marshal's 1st and 3rd statements will be true. Considering this, Ted's number of Postal Stamps will be 83, Marshal's will be 84 and Barney's will be 89.

Nature	Name	Number of Postal Stamps	Statement 1	Statement 2	Statement 3
Liar	Ted	83	F	F	F
Alternator	Marshal	84	T	F	T
Truth Teller	Barney	89	T	T	T

Case B: Marshal is a liar and Ted is an alternator. Now Barney's 3rd statement is true so Ted's 1st and 3rd statement becomes false. Regarding the number of Postal Stamps we can deduce that only for Barney i.e., 89, for Marshal we can say it can't be 84 and will be less than that of Ted's.

Nature	Name	Number of Postal Stamps	Statement 1	Statement 2	Statement 3
Alternator	Ted		F	T	F
Liar	Marshal	(<Ted's, not 84)	F	F	F
Truth Teller	Barney	89	T	T	T

The number of postal stamps with Barney is 89 when either Ted is a liar or Marshal is a liar.

[Bookmark](#)[FeedBack](#)

Sec 3

Q.51 [11594329]

The product of 1st five terms of an increasing AP is 29160. If the first, second and fourth terms of the AP are in GP, then what is the sum of the first 11 terms of the AP?

1 ☐ 2422 ☐ 1983 ☐ 2534 ☐ 275**Solution:****Correct Answer : 2** Answer key/Solution

Let the five terms of the AP be $(a - 2d)$, $(a - d)$, a , $(a + d)$ and $(a + 2d)$. Then,
 $(a - 2d)(a - d) \times a \times (a + d)(a + 2d) = 29160$... (i)
 Since the first, second and fourth terms of the AP are in GP.

$$\text{Therefore, } \frac{a-d}{a-2d} = \frac{a+d}{a-d}$$

$$\Rightarrow d(3d - a) = 0$$

$$\Rightarrow a = 3d$$

By putting $a = 3d$ in (i), we get

$$(3d - 2d)(3d - d) \times 3d \times (3d + d)(3d + 2d) = 29160$$

$$\Rightarrow d \times 2d \times 3d \times 4d \times 5d = 29160$$

$$\Rightarrow 120d^5 = 120 \times 3^5$$

$$\Rightarrow d = 3 \text{ and } a = 9$$

$$\text{Hence, the sum of the first 11 terms of the AP} = \frac{11}{2} [2 \times 3 + (11 - 1) \times 3] = 198.$$

Bookmark

FeedBack

Q.52 [11594329]

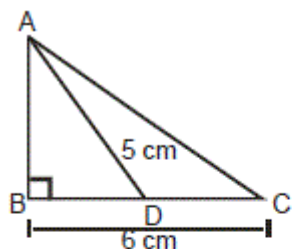
A right angled $\triangle ABC$ is such that $\angle ABC = 90^\circ$ and $BC = 6$ cm. D is a point on BC such that $AD = 5$ cm. If perimeter of $\triangle ABD =$ perimeter of $\triangle ADC$, then which of the following is true?

1 ☐ $\frac{CD}{BD} > 1$

2 ☐ $\frac{CD}{BD} < \frac{1}{5}$

3 ☐ $1 < \frac{CD}{BD} < 2$

4 ☐ $\frac{1}{5} < \frac{CD}{BD} < 1$

Solution:**Correct Answer : 4**[Answer key/Solution](#)

Perimeter of $\triangle ABD$ = Perimeter of $\triangle ADC$

$$\Rightarrow AB + BD + AD = AD + CD + AC$$

$$\Rightarrow AB + BD = CD + AC$$

Since $AC > AB$ (AC is hypotenuse)

$$\Rightarrow BD > CD$$

$$\Rightarrow \frac{CD}{BD} < 1$$

$AD = 5$ cm, therefore, $BD < 5$ cm.

Since $BD + CD = 6$ cm

Thus, $CD > 1$ cm

$$\text{Therefore, } \frac{CD}{BD} > \frac{1}{5}$$

Hence, the answer is $\frac{1}{5} < \frac{CD}{BD} < 1$.

Bookmark

FeedBack

Q.53 [11594329]

To do a piece of work, Z takes four times the time taken by X and Y together take to finish the work. X takes 5 times the time taken by Y and Z together. If X, Y, and Z together can finish the same work together in 12 days, then which one among the following options is false?

1 ☐ X and Z together are less efficient than Y.

2 ☐ Y is more efficient than Z.

3 ☐ X and Z together are more efficient than Y.

4 ☐ Z is more efficient than X.

Solution:**Correct Answer : 3****Time taken by Z : Time taken by (X + Y) = 4 : 1****So ratio of efficiency = 1 : 4****Time taken by X : Time taken by (Y + Z) = 5 : 1****So ratio of efficiency = 1 : 5****Let total work = 360 units****One day work of (X, Y, Z) together = 30 units****As efficiency is work done per day,****Thus work done per day by Z = 6 units****Work done per day by X and Y together = 24 units****Work done per day by X = 5****So work done per day by Y = 19 units.**

Bookmark

FeedBack

 Answer key/Solution**Q.54 [11594329]**

A is the smallest natural number such that $(A!)$ is divisible by each of the first fifteen prime numbers and B is the smallest natural number such that $(B!)$ is divisible by each of the first 8 prime numbers. Find the highest power of 2 in $[(A!) - (B!)]$.

1 ☐ 422 ☐ 163 ☐ 264 ☐ 47**Solution:****Correct Answer : 2****The first fifteen prime numbers is 47, therefore A = 47;****Similarly, the first 8 prime numbers is 19, therefore B = 19;** **$A! - B! = 47! - 19! = 19! \times \text{Odd number}$** **Hence, highest power of 2 in $(A! - B!)$ is same as the highest power of 2 in $19!$, which is 16.**

Bookmark

FeedBack

 Answer key/Solution**Q.55 [11594329]**

A boat travels downstream from Point A to B on a river. On its way back from B to A it travels one-fourth the distance in the time it covered the distance from A to B. What is the percentage increase in time taken to travel the distance AB by boat in still water.

1 ☐ 40%2 ☐ 25%3 ☐ 30%4 ☐ 60%**Solution:****Correct Answer : 4** Answer key/Solution

Let speed of boat be "b" and river be "r".

$$\frac{b+r}{b-r} = \frac{4}{1}$$

$$\Rightarrow b+r = 4b-4r$$

$$\Rightarrow 3b = 5r$$

$$\Rightarrow b = \frac{5}{3}r$$

$$\text{S(downstream): } b+r = \frac{5}{3}r+r = \frac{8r}{3}$$

For distance AB,

At $\frac{8}{3}r$ let time taken be 't' in still water speed will be $\frac{5r}{3}$.So drop in speed in by fraction $\frac{5}{8}$.So time increases by $\frac{8}{5}$.

Old time = t

$$\text{New time} = \frac{8}{5}t$$

$$\text{Increase in time} = \frac{3t}{5}$$

Hence, required percentage increase = 60%.

Bookmark

FeedBack

Q.56 [11594329]

Ram's incomes in year 1 and year 2 are in ratio of 1.25 : (1.333...). His incomes in year 2 and year 3 are in ratio of (3.666...) : 4.5. He receives a bonus at the end of year 3 which is 37.5% of his total income in 3 years. If his income in year 3 is Rs. 10,200 more than that in year 1, then find the bonus amount (in Rs.).

Solution:**Correct Answer : 41775**[Answer key/Solution](#)

Let Ram's income in year 1, year 2 and year 3 be I_1 , I_2 and I_3 respectively.

$$\therefore I_1 : I_2 = \left(1 + \frac{1}{4}\right) : \left(1 + \frac{1}{3}\right) = \frac{5}{4} : \frac{4}{3} = 15 : 16$$

$$\text{And, } I_2 : I_3 = \left(3 + \frac{2}{3}\right) : \left(4 + \frac{1}{2}\right) = \frac{11}{3} : \frac{9}{2} = 22 : 27$$

$$\therefore I_1 : I_2 : I_3 = 165 : 176 : 216$$

$$\text{Let } I_1 = 165x, I_2 = 176x, I_3 = 216x$$

$\therefore$ According to the question,

$$I_3 - I_1 = 10200$$

$$\Rightarrow 216x - 165x = 10200$$

$$\Rightarrow 51x = 10200$$

$$\Rightarrow x = 200$$

$$\text{Now, } (I_1 + I_2 + I_3) = 557 \times 200$$

$$\text{Hence, the bonus amount} = \frac{3}{8} \times 557 \times 200 = \text{Rs. } 41,775.$$

Bookmark

FeedBack

Q.57 [11594329]

When a container is 50% full it holds 60 litres less than when it is 35% empty. How many litres more does the container hold when it is 70% full than when it is 70% empty?

Solution:**Correct Answer : 160**[Answer key/Solution](#)

Let the volume of the container be V litres.

35% empty means 65% full.

$\therefore$ According to the question,

$$\frac{65}{100}V - \frac{50}{100}V = 60$$

$$\Rightarrow 15V = 60 \times 100$$

$$\Rightarrow V = 400 \text{ litres.}$$

$$\text{Now, when the container is 70\% full, it holds} = \frac{70}{100} \times 400 = 280 \text{ litres.}$$

$$\text{And, 70\% empty means 30\% full, so, it holds} = \frac{30}{100} \times 400 = 120 \text{ litres}$$

Hence, the container holds $(280 - 120) = 160$ litres more when it is 70% full than when it is 70% empty.

Bookmark

FeedBack

Q.58 [11594329]

$$S_1 = 2 + 5 + 8 + 11 + \dots + 302$$

$$S_2 = 4 + 9 + 14 + 19 + \dots + 504$$

Find the sum of all the terms which are not common to the above two series.

1 ☐ 34746

2 ☐ 6260

3 ☐ 37876

4 ☐ None of these

Solution:

Correct Answer : 1

[Answer key/Solution](#)

$$S_1 = 2 + 5 + 8 + 11 + \dots + 302$$

$$S_1 = \frac{101}{2} [2 + 302] = 15352$$

$$S_2 = 4 + 9 + 14 + 19 + \dots + 504$$

$$S_2 = \frac{101}{2} [4 + 504] = 25654$$

Terms which are common to the 2 series; 14, 29, 44, 59, ..., 299;

$$\text{and their sum is } = \frac{20}{2} [14 + 299] = 3130$$

$$\text{Sum of terms which are not common} = (15352 + 25654) - \underbrace{2 \times 3130}$$

(Since common terms are added twice, (in both series))

$$= 34746.$$

Bookmark

FeedBack

Q.59 [11594329]

PQRS is a parallelogram, where $PQ > QR$. Point A on PQ divides it in the ratio of 7 : 3 and point B on RS divides it in the ratio of 7 : 3. If AR meets QS at X and BP meets QS at Y such that $AR \parallel BP$, then find the area of the circle formed by taking XY as its diameter. [Given $QS = 26$ units and take

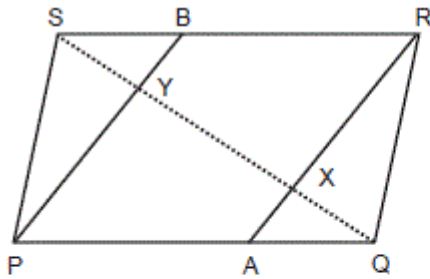
$$\pi = \frac{22}{7}]$$

1 ☐ 120 sq. units

2 ☐ 144 sq. units

3 ☐ 156 sq. units

4 ☐ 154 sq. units

Solution:**Correct Answer : 4**[Answer key/Solution](#)
 $\Delta AXQ \sim \Delta PYQ$ and $\Delta SYB \sim \Delta SXR$

$$\frac{AQ}{AP} = \frac{QX}{XY} = \frac{3}{7} \quad \frac{SB}{BR} = \frac{SY}{XY} = \frac{3}{7}$$

Combining, $XQ : XY : SY = 3 : 7 : 3$ as $QS = 26$ units, $XY = 14$ units

$$\text{Area of circle with } XY \text{ as diameter} = \pi \left(\frac{XY}{2} \right)^2 = \frac{22}{7} \times 7 \times 7 = 154 \text{ sq. units.}$$

Bookmark

FeedBack

Q.60 [11594329]

In a convex 16 sided polygon, the measures of all interior angles are in an increasing arithmetic sequence with integral values. Find the measure of the largest angle of the polygon (in degrees).

1 ☐ 1652 ☐ 1553 ☐ 1754 ☐ 155

Solution:**Correct Answer : 1** Answer key/SolutionLet the smallest angle = a Common differences = d Thus, 2nd smallest angle = $a + d$ and so on till the largest angle = $a + 15d$ Sum of interior angles = $(n - 2) \times 180 = 14 \times 180$

$$\frac{16}{2}[2a + (16 - 1)d] = 14 \times 180$$

$$\Rightarrow 2a + 15d = 315$$

$$\Rightarrow a + (a + 15d) = 315$$

Now ' a ' is the smallest angle and ' $a + 15d$ ' is the largest angle.As it is a convex polygon, all angles have to be less than 180° .If $a + 15d < 180$ and sum of $a + (a + 15d) = 315$ Also, $a > 135$ (Otherwise largest angle will become greater than 180°) $a + 15d < 180$ or, $15d < 45$ As both ' a ' and ' d ' have to be positive integers.Value of $d = 1$ or 2 .When $d = 1 \Rightarrow 2a + 15 = 315$

$$2a = 300$$

$$\Rightarrow a = 150$$

When $d = 2 \Rightarrow 2a + 30 = 315$

$$2a = 285$$

$$a \neq 142.5$$

Thus largest angle = $150 + 15 \times 1 = 165^\circ$.

Bookmark

FeedBack

Q.61 [11594329]If $\log_2(\log_3(\log_4(x^2 + 4x + 4))) = 0$, then find the sum of all values of x .1 ☐ 162 ☐ 43 ☐ -164 ☐ -4

Solution:**Correct Answer : 4****Your Answer : 4** Answer key/SolutionWe know that $\text{Log}_2(\text{log}_3(\text{log}_4(x^2 + 4x + 4))) = 0$ Therefore, $(\text{log}_3(\text{log}_4(x^2 + 4x + 4))) = 1$ $\Rightarrow \text{log}_4(x^2 + 4x + 4) = 3$ $\Rightarrow (x^2 + 4x + 4) = 64$ $(x^2 + 4x - 60) = 0$ Therefore, $x = 6, -10$

Required answer = -4.

Bookmark

FeedBack

Q.62 [11594329]Let $S_1 = A_1 A_2 A_3 \dots A_{49}$ and $S_2 = \frac{1}{B_4 B_5 B_6 \dots B_{51}}$, where $A_n = \frac{2}{n} + 1$ and $B_n = 1 - \frac{2}{n}$,then what is the value of $\frac{S_1}{S_2}$?1 ☐ 2:12 ☐ 1:33 ☐ 3:14 ☐ 2:3**Solution:****Correct Answer : 3** Answer key/Solution

$$S_1 = 3 \times 2 \times \frac{5}{3} \times \frac{6}{4} \times \frac{7}{5} \times \dots \times \frac{51}{49}$$

$$= \frac{3 \times 2 \times 50 \times 51}{3 \times 4} = 25 \times 51$$

$$S_2 = \frac{1}{\frac{2}{4} \times \frac{3}{5} \times \frac{4}{6} \times \dots \times \frac{49}{51}}$$

$$= \frac{1}{\frac{2 \times 3}{50 \times 51}} = \frac{50 \times 51}{2 \times 3} = \frac{51 \times 25}{3}$$

$$\therefore \frac{S_1}{S_2} = \frac{25 \times 51}{\frac{51 \times 25}{3}} = 3$$

Bookmark

FeedBack

Q.63 [11594329]

How many values can natural number 'n' take, if n! is a multiple of 5^9 but not 5^{13} ?

1 ☐ 14

2 ☐ 15

3 ☐ 16

4 ☐ 17

Solution:

Correct Answer : 2

Smallest value of n! which is a multiple of 51 will be 51.

Now 40! has 9 powers of 5 so it is a multiple of 5^9 .

55! has 13 powers of 5.

So all numbers from 40 to 54 can be the value of x. i.e., 15 values.

Bookmark

FeedBack

 Answer key/Solution

Q.64 [11594329]

Catherine overtakes a car travelling from Delhi to Lucknow at 6:30 PM. Catherine reaches Lucknow at 8:00 PM. Immediately after reaching Lucknow, Catherine starts her journey back and crossed the same car at 8:30 PM, which was still moving towards Lucknow. If both Catherine and the car are travelling at constant speeds, at what time will the car reach Lucknow?

1 ☐ 9:00 PM

2 ☐ 9:15 PM

3 ☐ 9:30 PM

4 ☐ 9:45 PM

Solution:

Correct Answer : 3

Rather than making equations, let us solve it mentally. Let us say that Catherine travels x km to reach Lucknow (after overtaking the car). Then she travelled this distance in 1.5 hours. Now, it travelled back to meet the car in .5 hours. This time she must have travelled a distance of $x/3$ km. During this time, the car must have travelled the difference of these two distances i.e., $2x/3$ km in two hours. So, according to unitary method, it will travel "x" km in 3 hours. Therefore, the car will reach Lucknow at 9:30 PM.

Bookmark

FeedBack

 Answer key/Solution

Q.65 [11594329]

If $\frac{x}{112} = \frac{343}{\sqrt[3]{y}}$ and $\frac{y}{80} = \frac{125}{\sqrt[3]{x}}$, then what is the value of $\left(\frac{x}{y}\right)^{\frac{3}{2}}$?

1 ☐ $\left(\frac{7}{5}\right)^{\frac{3}{2}}$

2 ☐ $\left(\frac{5}{7}\right)^{-9}$

3 ☐ $\left(\frac{7}{5}\right)^{-6}$

4 ☐ $\left(\frac{5}{7}\right)^9$



Solution:

Correct Answer : 2

Your Answer : 2

Given:

$$\frac{x}{112} = \frac{343}{\sqrt[3]{y}} \text{ and } \frac{y}{80} = \frac{125}{\sqrt[3]{x}}$$

$$\Rightarrow \frac{x\sqrt[3]{y}}{y\sqrt[3]{x}} = \frac{343 \times 112}{125 \times 80} \Rightarrow \left(\frac{x}{y}\right)^{\frac{2}{3}} = \frac{343 \times 7}{125 \times 5} = \left(\frac{7}{5}\right)^4$$

$$\Rightarrow \left(\frac{x}{y}\right)^{\frac{2}{3} \times \frac{3}{2}} = \left(\frac{7}{5}\right)^{4 \times \frac{3}{2}} = \left(\frac{7}{5}\right)^6 \Rightarrow \left(\frac{x}{y}\right)^{\frac{3}{2}} = \left(\frac{7}{5}\right)^{6 \times \frac{2}{3}} = \left(\frac{7}{5}\right)^9 \text{ or } \left(\frac{5}{7}\right)^{-9}$$

Bookmark

FeedBack

[Answer key/Solution](#)

Q.66 [11594329]

In year 2015, population on an island Nutas was observed to have increased by 10% over the previous year. In 2016 the population increased by 20% over the previous year and in 2017 it increased by 10% over the previous year. However in 2018, the population recorded was same as in year 2017. If at the end of year 2018, the population is 12204 more than the population recorded at the end of 2014, then what is the population at the end of 2014?

**Solution:****Correct Answer : 27000****Your Answer : 27000**
[🔍 Answer key/Solution](#)

Let x be the population on the island at the end of 2014.

$x \left[1 + \frac{10}{100} \right] \left[1 + \frac{20}{100} \right] \left[1 + \frac{10}{100} \right]$ is the population recorded in year 2018.

$$x \left(\frac{11}{10} \times \frac{12}{10} \times \frac{11}{10} \right) - x = 12204$$

$$\Rightarrow x \left[\frac{1452 - 1000}{1000} \right] = 12204$$

$$\Rightarrow x = \frac{12204 \times 100}{452} = 27000.$$

[Bookmark](#)
[FeedBack](#)
Q.67 [11594329]

If $\frac{1}{x} - \frac{1}{y} = \frac{1}{7}$ and $xy^2 - yx^2 = 448$, then find the value of $y - x$, where $y > x$ and x, y are the positive real numbers.

**Solution:****Correct Answer : 8****Your Answer : 8**
[🔍 Answer key/Solution](#)

Given:

$$\frac{1}{x} - \frac{1}{y} = \frac{1}{7} \text{ and } xy^2 - y^2x = 448$$

$$\Rightarrow \frac{xy^2 - y^2x}{x^2y^2} = \frac{448}{x^2y^2} \Rightarrow \frac{1}{x} - \frac{1}{y} = \frac{448}{(xy)^2} \Rightarrow \frac{1}{7} = \frac{448}{(xy)^2}$$

$$\Rightarrow (xy)^2 = 7 \times 448 = 49 \times 64$$

$$\Rightarrow xy = 7 \times 8$$

$$\text{Now, } \frac{1}{x} - \frac{1}{y} = \frac{y-x}{xy} = \frac{1}{7} \Rightarrow y-x = \frac{7 \times 8}{7} = 8.$$

[Bookmark](#)
[FeedBack](#)
Q.68 [11594329]

Monica is making a cocktail by mixing drinks A, B, and C in ratio 3 : 5 : 8. She further adds A and B till the ratio of A, B, and C in the cocktail becomes 8 : 5 : 3. Find the ratio of amounts of A and B added.

1 ☐ 7 : 52 ☐ 10 : 33 ☐ 11 : 54 ☐ 8 : 3**Solution:****Correct Answer : 3****Your Answer : 3**
[🔍 Answer key/Solution](#)

Let the quantity of A, B and C used is 30 L, 50 L and 80 L.

Now, we require ratio to be 8 : 5 : 3 by adding only A and B, keeping C constant.

$$\text{So, quantity of A} = \frac{8}{3} \times 80 = \frac{640}{3} \text{ L}$$

$$\text{Quantity of B} = \frac{5}{3} \times 80 = \frac{400}{3} \text{ L}$$

$$\text{Thus quantity of added} = \frac{640}{3} - 30 = \frac{550}{3} \text{ L}$$

$$\text{Quantity of B added} = \frac{400}{3} - 50 = \frac{250}{3} \text{ L}$$

$$\text{Ratio of A and B} = \frac{550}{3} : \frac{250}{3} = 11 : 5.$$

[Bookmark](#)
[FeedBack](#)
Q.69 [11594329]

A function is defined as $g(n) = g(n - 1) + n - 1$ for integral values of n greater than 1. If $g(1) = 1$, then find $g(25)$.

1 ☐ 892 ☐ 1333 ☐ 2754 ☐ 301

Solution:**Correct Answer : 4****Your Answer : 4** Answer key/Solution

$$g(n) = g(n-1) + n - 1, g(1) = 1$$

$$\text{For } n = 2, g(2) = g(1) + 2 - 1 = 2$$

$$n = 3, g(3) = g(2) + 3 - 1 = 4$$

$$n = 4, g(4) = g(3) + 4 - 1 = 7$$

$$\vdots \quad \quad \quad \vdots$$

$\therefore$ The series is 1, 2, 4, 7, 11, ... so on.

Now, we know the series

1, (1 + 2), (1 + 2 + 3), ... is 1, 3, 6, 10, ...

If we compare above two series, if we add 1 to second series, it is equivalent to the first series from second term.

$$\text{Hence, } g(25) = \left(\frac{24 \times 25}{2} \right) + 1 = 301.$$

Bookmark

FeedBack

Q.70 [11594329]

A shopkeeper gives 3 DVDs free with the purchase of every 7 TVs and 4 TVs free with the purchase of every 9 DVDs. Ross purchased 18 DVDs and Rachael purchased 21 TVs. The percentage discount obtained by Ross on his entire purchase was 25%. What was the percentage discount obtained by Rachael on her entire purchase? (Assume the price of all DVDs and price of all TVs is same.)

1 ☐ 50%2 ☐ 25%3 ☐ 36.36...%4 ☐ 37.5%**Solution:****Correct Answer : 3** Answer key/Solution

Let the price of each T.V. be x and that of each DVD be y .

Ross purchased 18 DVDs so, he must have got 8 TVs free.

Had he got no discount, then the value of the marked price would have become = $18y + 8x$

Now, according to the question,

$$\frac{8x}{18y + 8x} = \frac{1}{4} \quad (\text{As discount} = 25\%)$$

$$\Rightarrow 4x = 3y$$

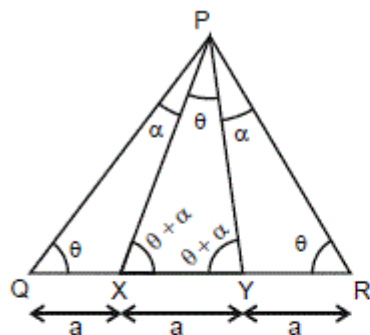
$$\therefore \text{Percentage discount obtained by Rachel} = \frac{9y}{21x + 9y} = \frac{9 \times \frac{4x}{3}}{21x + 9 \times \frac{4x}{3}} = \frac{4}{11} = 36.36\%.$$

Bookmark

FeedBack

Q.71 [11594329]

In an isosceles triangle PQR with $PQ = PR$, PX and PY trisect QR such that $QX = XY = YR$. If $\angle XPY$ is equal to $\angle PQR$ and the area of triangle PQR is $\frac{(75\sqrt{7})}{4}$, then find PR.

1 ☐ 102 ☐ 143 ☐ 164 ☐ 12**Solution:****Correct Answer : 1**[Answer key/Solution](#)

By symmetry,

$$\angle QPX = \angle RPY = \alpha$$

$$\angle PXY = \angle QPX + \angle PQX \quad [\text{Exterior angle}]$$

Thus in $\triangle PRX$,

$$\angle RPX = \angle RXP = Q + \alpha$$

$$\text{So, } PR = RX = 2a$$

$$\text{Height of isosceles } \triangle PQR \text{ on base } QR = \sqrt{(2a)^2 - \left(\frac{3a}{2}\right)^2} = \sqrt{\frac{7a^2}{4}} = \frac{a\sqrt{7}}{2}$$

$$\text{Area} = \frac{1}{2} \times 3a \times \frac{a\sqrt{7}}{2} = \frac{75\sqrt{7}}{4}$$

$$\Rightarrow a^2 = 25$$

$$\Rightarrow a = 5$$

$$\text{Hence, } PR = 2a = 10.$$

Bookmark

FeedBack

Q.72 [11594329]

If the value of $\frac{3}{2\log_2 72^4} + \frac{1}{\log_3 72^4}$ can be expressed as $\frac{a^3}{b^3}$, where a and b are co-prime numbers, then find the value of (a + b).

Solution:**Correct Answer : 3**

$$\begin{aligned}\frac{3}{2\log_2 72^4} + \frac{1}{\log_3 72^4} &= \frac{3}{2 \times 4\log_2 72} + \frac{1}{4\log_3 72} \\ &= \left(\frac{3}{8}\right)\log_{72} 2 + \left(\frac{1}{4}\right)\log_{72} 3 \\ &= \log_{72} 2^{\left(\frac{3}{8}\right)} + \log_{72} 3^{\left(\frac{1}{4}\right)} = \log_{72} 2^{\left(\frac{3}{8}\right)} \cdot 3^{\left(\frac{1}{4}\right)} \\ &= \log_{72} 8^{\left(\frac{1}{8}\right)} \cdot 3^{\left(\frac{2}{8}\right)} = \log_{72} 8^{\frac{1}{8}} \cdot 9^{\frac{1}{8}} = \log_{72} 72^{\left(\frac{1}{8}\right)} = \frac{1}{8}.\end{aligned}$$

$$\therefore \frac{a^3}{b^3} = \frac{1}{8}$$

$$\therefore a = 1 \text{ and } b = 2$$

$$\text{So, } a + b = 1 + 2 = 3.$$

Bookmark

FeedBack

 Answer key/Solution**Q.73 [11594329]**

Given that x and y are natural numbers such that $2x + 5y = 120$. If $|x - y| \geq 10$, how many ordered pairs (x, y) are possible?

Solution:**Correct Answer : 9**

$$2x + 5y = 120$$

where x, y are natural numbers.

$$\begin{array}{l} \text{Possible values of } x \text{ \& } y \text{ are: } \left. \begin{array}{ll} x = 5 & y = 22 \\ = 10 & = 20 \\ = 15 & = 18 \\ = 20 & = 16 \\ = 25 & = 14 \\ = 30 & = 12 \\ = 35 & = 10 \\ = 40 & = 8 \\ = 45 & = 6 \\ = 50 & = 4 \\ = 55 & = 2 \end{array} \right\} \end{array}$$

only for these values of x & y, we have $|x - y| \geq 10$.

Hence, only 9 pairs of acceptable values.

Bookmark

FeedBack

 Answer key/Solution

Q.74 [11594329]

In an organisation, $\frac{2}{7}$ th of the employees are men and $\frac{1}{5}$ th of the men are married.

Further, $\frac{1}{2}$ of all the married men have at least 1 child. $\frac{3}{4}$ th of all the women employees are married and $\frac{1}{3}$ rd of all the married women have at least 1 child. What could be the minimum number of employees in the organization?

Solution:**Correct Answer : 140**[Answer key/Solution](#)

Let the number of employees be x .

$$\text{Number of men} = \frac{2}{7}x; \quad \text{Number of women} = \frac{5}{7}x$$

$$\text{Number of married men} = \frac{1}{5}\left(\frac{2}{7}x\right) \text{ and number of married women} = \frac{3}{4}\left(\frac{5}{7}x\right)$$

$$\text{Number of married men with atleast 1 child} = \frac{1}{2} \times \frac{1}{5} \times \frac{2}{7}x = \frac{x}{35}$$

$$\text{Number of married women with atleast 1 child} = \frac{1}{3} \times \frac{3}{4} \times \frac{5}{7}x = \frac{5}{28}x$$

Therefore, x should be a multiple of 35 and 28.

Hence, minimum value of $x = \text{LCM}(35, 28) = 140$.

[Bookmark](#)[FeedBack](#)**Q.75 [11594329]**

Fifteen friends went for a dinner at a restaurant. While paying the bill, 10 friends paid Rs. 30 each whereas each of the remaining 5 friends paid 1.5 times the average bill amount. Find the average bill amount (in Rs.).



[Q. Answer key/Solution](#)**Solution:****Correct Answer : 40****Your Answer : 40****Let x be the average bill amount per person.****Total amount paid = $15x$** **10 friends paid a total of Rs. 300.****Remaining friends paid $1.5x$ each.****Now, $300 + 5(1.5x) = 15x$** **Therefore, $7.5x = 300$** **Therefore, required answer is Rs. 40.**[Bookmark](#)[FeedBack](#)